

Energy market liberalization and the emergence of new energy ventures in Germany

Prashant Pant^{a,*} , Frank-Martin Belz^b 

^a Center for Combined Smart Energy Systems (CoSES), Technical University of Munich, Munich, Germany

^b Chair of Corporate Sustainability, Technical University of Munich, Munich, Germany

HIGHLIGHTS

- Analysis over 25 years suggests a positive association between German energy policies and the emergence of new ventures.
- New entrepreneurial ventures are observed to be highly concentrated in downstream, customer-centric market segments.
- The findings indicate that key growth areas currently center on BESS, heat pumps, and smart energy management solutions.
- The market adoption of available technologies appears to be heavily reliant on policy incentives that influence consumer behavior.

ARTICLE INFO

Keywords:

Energy market liberalization
Energy ventures
Germany
New business models
Policy-driven entrepreneurship
Renewable energy integration

ABSTRACT

This paper investigates how energy market liberalization and the policies supporting renewable energy integration have influenced the emergence of new energy ventures in Germany. A wide range of policies is examined, including the Energy Industry Act (EnWG), the Renewable Energy Sources Act (EEG), the Buildings Energy Act (GEG), and several others that have shaped the sector. The analysis is organized around the electricity value chain by considering upstream, midstream, and downstream segments and traces policy impacts on venture creation through data and case studies. The findings suggest a noticeable trend between regulatory measures and the development of innovative business models. Importantly, the study highlights that technological advancement alone is not sufficient; policy support, financial incentives, and market access mechanisms are critical drivers of both entrepreneurial activity and new venture emergence. The study provides practical insights for policy-makers internationally, particularly in countries in the Global South that have yet to undergo energy market liberalization.

1. Introduction

Energy market liberalization aims to dismantle monopolistic structures, promote competition, and potentially lower end-user prices (Lenz et al., 2019; Fatras et al., 2022; Rotaru, 2013). Nevertheless, electricity prices in Germany remain among the highest in Europe (Kappner et al., 2023), reflecting a combination of factors, including market design, infrastructure costs, and energy policy decisions (Liebensteiner et al., 2025). Germany began this transition in April 1998 (Rotaru, 2013), introducing unbundling of generation, transmission, distribution, and retail functions, as well as balancing service providers (BSPs) to support Transmission System Operators (TSOs) and ensure grid stability

(Pant et al., 2024). Transmission and distribution grids continue to operate as regulated natural monopolies, with the Federal Network Agency (BNetzA) overseeing access and fees (Rotaru, 2013). Fig. 1 illustrates the key stakeholders in the liberalized German electricity market.

Germany's energy market liberalization was closely aligned with strategic policy interventions aimed at expanding the role of renewable energy within its power system. This coordinated approach led to a profound transformation of the energy landscape: the share of renewables in gross electricity consumption increased from 6.3 % in 2000 to 54.4 % by 2024 (Burger, 2024). Central to this shift was the rise of small-scale prosumers and decentralized generators. The interplay between liberalized market structures and supportive renewable energy policies proved

* Corresponding author.

Email address: prashant.pant@tum.de (P. Pant).

Nomenclature			
AbLaV	Regulation on Interruptible Loads	EV	Electric Vehicle
aFRR	Automatic Frequency Restoration Reserve	EPEX SPOT	European Power Exchange Spot Market
BDEW	German Association of Energy and Water Industries	FCR	Frequency Containment Reserve
BESS	Battery Energy Storage Systems	FiT	Feed-in Tariff
BNetzA	Federal Network Agency	GDEW	Digitization of the Energy Transition Act
DER	Distributed Energy Resources	mFRR	Manual Frequency Restoration Reserve
DERMS	Distributed Energy Resource Management Systems	PICASSO	Platform for International Coordination of Automated Frequency Restoration
DSO	Distribution System Operator	PV	Photovoltaic
EBGL	Electricity Balancing Guideline	SMGW	Smart Meter Gateway
EEG	Renewable Energy Sources Act	SOGL	System Operation Guideline
EnWG	Energy Industry Act	TSO	Transmission System Operator
		VPP	Virtual Power Plant

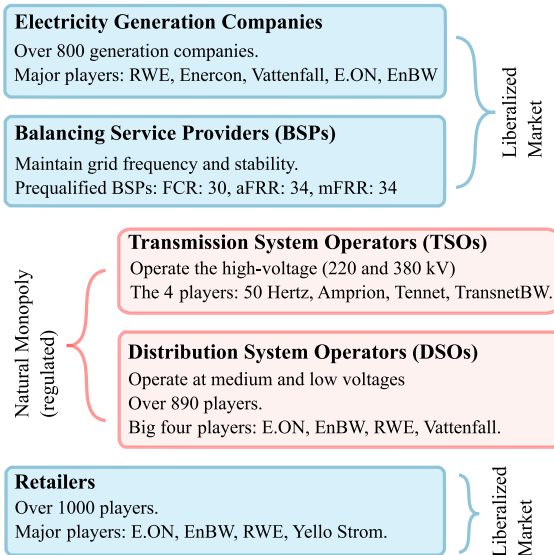


Fig. 1. Structure of the German electricity market, from generation to retailers.

instrumental in cultivating a dynamic ecosystem of energy ventures (Bendig et al., 2025).

Today, this ecosystem encompasses a wide array of actors, including independent power producers, ancillary service providers, aggregators, virtual power plants, arbitrageurs, portfolio managers, project developers, innovative retailers, and technology firms offering advanced hardware and software solutions. Assessing how these intertwined policy and market frameworks have shaped the growth and success of such ventures is critical for informing similar transitions in other countries. This paper explores the key historical and policy milestones that influenced the development of energy ventures in Germany, categorizes these ventures within the electricity value chain and market structure, and analyzes the drivers of their success, with the aim of extracting insights applicable to broader energy transitions.

2. Literature review

The liberalization of the European electricity market began over three decades ago (Rotaru, 2013). Although most EU member states have formally liberalized their markets and participate in platforms such as Nord Pool, EPEX SPOT, and the European Energy Exchange (EEX), a fully unified internal electricity market in Europe remains the ultimate goal (Rotaru, 2013). Research on electricity market liberalization and the transition toward renewable-dominated power systems generally falls into two main streams.

The first, which examines the institutional and economic outcomes of the liberalization framework, has been studied extensively (Lenz et al., 2019; Schmid-Petri and Elschner, 2024; Pepermans, 2019). This research typically assesses aspects such as market design, the effectiveness of unbundling generation from transmission, and the performance of key actors like TSOs and DSOs in their regulated roles. The second stream explores the impact of market liberalization and public policies on innovation and the emergence of new energy ventures. While this field has grown significantly, much of the research focuses on specific aspects, such as public R&D funding's effect on patent activity (Plank and Dobliger, 2018), the growth of citizen energy cooperatives (Ahlemeyer et al., 2022), or macroeconomic impacts (Sievers et al., 2019). Consequently, a comprehensive analysis that systematically links a broad range of policy interventions to the emergence of diverse business models across the entire electricity value chain has yet to be fully explored. This paper aims to fill that specific gap, a task that is crucial for several key reasons:

- Helps anticipate energy market trends, investment opportunities, and regulatory challenges.
- Supports the development of effective policies for accelerating the renewable energy transition.
- Highlights how liberalization fosters innovative energy ventures and entrepreneurial activity.

To identify relevant sources, a systematic search of academic databases (Scopus, Web of Science, and Google Scholar) was conducted, covering the period from 2000 to 2025, complemented by a review of relevant grey literature (e.g., market reports, technical reports, company websites, and white papers). The search strategy combined keywords across three core areas: (i) *Policy and Regulation* (e.g., “energy market liberalization”, “renewable energy policy”, “feed-in tariff”, “EEG”, “Energiewende”, and other relevant policies), (ii) *Entrepreneurial Outcomes* (e.g., “energy ventures”, “startups”, “new business models”, “innovation”), and (iii) *Geographic Context* (e.g., “Germany”, “Europe”). Included sources were required to be peer-reviewed or credible industry reports and to focus on the post-1998 liberalization period. Selection prioritized works that explicitly linked policy interventions to firm-level entrepreneurial or innovation outcomes, while excluding studies that were purely technical or macroeconomic modeling without addressing firm dynamics. Reference lists of relevant publications were also examined to identify additional relevant works.

The role of social innovation and sustainable value creation resulting from the energy transition has been examined by Mihailova (2023), though the discussion remains limited and lacks emphasis on practical implications. A cross-country review by Nicolli and Vona (2019), finds that energy market liberalization positively influences policies aimed at increasing renewable energy adoption across OECD countries.

Table 1

Peer-reviewed and academic studies on analyzing the impact of energy policies and renewable energy integration on the creation of energy ventures.

Reference (Year)	Policy Focus	Methodology	Key Findings	Limitations
Bürer and Wüstenhagen (2009)	Renewable energy policy (feed-in tariffs vs. quotas).	Survey of 60 venture investors (Europe/North America) with qualitative interviews.	Feed-in tariffs (German EEG) are perceived as the most effective policy for attracting clean-energy investment; quotas/auctions rank significantly lower in investor preference.	Small, non-representative sample; reflects investor perceptions (not actual venture data); multi-country context, dated (pre-2010).
Haelg (2017)	Deployment policies for solar PV innovation (Germany, EU).	Based on mixed methods: patent analysis, interviews, policy analysis.	Shows deployment policies (e.g., feed-in tariffs) significantly drove innovation in solar PV; early support critical for technology cost reduction and firm emergence.	Focused on solar PV only; historical scope (up to 2015); Germany used as a main case but includes broader international context.
Böhringer et al. (2017)	Renewable subsidy impacts on income distribution	CGE model with microsimulation	Subsidies raise electricity prices regressively; low-income households are disproportionately affected.	No focus on innovation or venture creation; model-based with assumed parameters
Aniello and Többen (2019)	German EEG—effect on North Rhine-Westphalia (NRW) economy.	Regional computable general equilibrium (CGE) model using NRW-specific input–output economic data to simulate policy impacts.	EEG-driven deployment of renewables (2011) led to a net loss of ≈ €2.35 billion in value added nationally, with NRW experiencing ≈40 % of the negative impact; local production of renewables provided smaller positive effects.	Single-year (2011) snapshot; static model assumptions; regional focus (NRW) limits generalization; no direct link to firm entry data.
Sievers et al. (2019)	Macroeconomic impacts of German energy transition (renewables and efficiency).	Macroeconomic modeling with regional, sectoral detail.	Positive net impacts on economic growth and employment; regional benefits stronger in north and east Germany; construction and electricity generation sectors grow; lignite mining declines.	Focus on macroeconomic impacts; no direct link to venture formation; model-based projections.
Singh et al. (2021)	German Energiewende context (renewable energy startups).	Data-driven (web-scraped) analysis of ≈3200 German energy startups – clustering by technology and business model.	Identifies major tech/business-model clusters (e.g., storage, smart-grid, energy apps) among startups; finds spatial concentrations reflecting existing industrial hubs.	Descriptive, not causal; relies only on online data completeness; does not quantify policy effects on venture rates; only limited ventures; pre-2020 study.
Ahlemeyer et al. (2022)	EEG-driven citizen energy cooperatives (Germany).	Qualitative review + 12 interviews with cooperative managers.	Highlights critical success factors for energy cooperatives (simple business models, committed leaders, social mission) and notes that existing regulatory conditions often hinder new projects.	Focus on cooperatives only (not all venture types); small, regional sample; findings are qualitative (no quantitative measure of policy effect); cooperative-specific context.
Xia-Bauer et al. (2022)	Business models for renewable energy prosumers (Germany).	Case study analysis and policy mapping.	Identifies innovative business models (e.g., peer-to-peer trading, energy communities) that empower prosumers under EEG framework; emphasizes the role of intermediaries and digital platforms.	Limited generalizability; mostly conceptual analysis; early-stage models; lacks quantitative evaluation of market impact.
Muhammad and Hoffmann (2024)	Role of green finance and innovation in German energy transition.	Empirical econometric analysis using panel data on green investments, patents, and energy performance indicators.	Finds strong correlation between green financial flows, clean-tech patenting, and CO ₂ reduction; green finance and tech innovation jointly accelerate energy transition.	Correlational (not causal); firm-level dynamics not isolated; limited exploration of specific business models or policy interactions.
Bendig et al. (2025)	Renewable energy and impact on entrepreneurship (international).	Systematic literature review (142 articles) of renewables entrepreneurship research.	Categorizes types of renewable energy ventures and their sustainability impacts; identifies numerous success factors and emphasizes the need for supportive policy frameworks to promote startups.	Broad scope (not Germany-specific); literature-based (no new data); emphasis on tech adoption (transformative models understudied); does not isolate specific policy impacts.

Complementing this, work from [Plank and Dobliger \(2018\)](#) provides firm-level evidence from the German renewable energy sector, showing that public R&D funding, driven by supportive policy frameworks, significantly enhances innovation output, particularly in terms of patent activity.

Another study ([Bendig et al., 2025](#)), examines the growth of renewable energy-based entrepreneurial ventures, although it focuses primarily on the generation segment. Solar and wind power plants are attracting significant investment, driven by the global consensus on the climate risks associated with fossil fuels ([Müller and Sacco, 2024](#)). However, power generation is only one element of liberalized electricity markets; the renewable energy integration has created opportunities for ventures across multiple levels of the energy market chain, highlighting the need for a comprehensive interdisciplinary study across the electricity supply chain, an aspect less explored in existing literature.

Reports from [McKinsey \(2022\)](#) and the World Economic Forum ([World Economic Forum, 2025](#)) have acknowledged the business potential arising from the ongoing energy transition. However, they do

not explicitly explore the full range of possibilities, nor do they affirm the positive correlation between the energy market liberalization and the creation of new energy ventures. Similarly, [Facchinetti \(2022\)](#) presented several business opportunities emerging from the energy transition, but only a limited number of possibilities are addressed, and they are not linked to the policies that could enable their practical realization. Establishing this link is crucial, as the theoretical potential and the conditions for real-world implementation are distinct and require separate consideration. Similarly, [Mouritz et al. \(2012\)](#) highlight energy trading, participation in ancillary service markets, and microgrid business models as possible business opportunities emerging from the energy transition.

The energy entrepreneurship ecosystem in Germany is rapidly growing. The Start Up Energy Transition (SET) initiative, organized by the German Energy Agency (dena) and the World Energy Council, seeks to drive and bring out innovation and practical entrepreneurial potential in the energy transition ([Start Up Energy Transition Initiative, 2024](#)). Since 2017, the SET100 list has annually recognized the 100 most promising

global energy startups across categories like clean energy generation, energy storage and distribution, mobility solutions, and energy efficiency. Another study from 2020, Singh et al. (2021) conducted a data-driven study to find the correlation between energy transition and the emergence of business models around the energy ecosystem; however, the energy start-up ecosystem is quite dynamic and has evolved since then due to new policies, innovations, and geopolitical conditions.

A cause-and-effect analysis of the German energy transition is presented in Kappner et al. (2023) to examine how various societal and political changes have influenced the transition. The study finds that technologies create opportunities for major changes and improvements, but do not initiate these changes themselves. Instead, technological advancements primarily respond to needs driven by new regulations and economic developments. A clear example of this dynamic is the emergence of Virtual Power Plants (VPPs), as developed by Sonnen GmbH and Next Kraftwerke (Mao et al., 2020; E. S. News, 2023; Birk et al., 2021). Regulatory changes allowing decentralized systems to participate in ancillary service markets enabled Sonnen to aggregate residential battery storage systems into a certified VPP. This innovation was not driven purely by technological capability, but by new market opportunities created through political and economic reforms aimed at decentralizing and stabilizing the energy system.

2.1. Identified research gaps

The literature review of peer-reviewed academic sources has been conducted and analyzed according to their policy focus, methodology, key findings, and limitations. The synthesized overview of these studies is presented in the Table 1. Despite the growing body of work on renewable energy policy and entrepreneurship, several important research gaps remain unaddressed. These include:

1. How has energy market liberalization influenced the emergence of new energy ventures, particularly across different segments of the electricity sector, and how have these segments been uniquely impacted?
2. How have specific policies facilitating market liberalization and renewable energy integration affected the growth of energy ventures in different segments of the electricity value chain?

2.2. Paper contribution

This paper aims to provide the following contributions:

1. Analyze the hypothesis that energy market liberalization and policies supporting renewable energy integration have led to the emergence of new energy ventures, and identify and categorize the segments where business opportunities have emerged.
2. Examine how policy interventions such as the EnWG, EEG, BEG, EnWG-14a, and several others have influenced the founding of energy ventures in the electricity sector through a cause-effect analysis, supported by detailed case study examples.

The paper is structured as follows: First, we review the literature and related works. Second, we provide an overview of the most significant policy changes regarding energy market liberalization in Germany. Then, we analyze a dataset from the German Association of Energy and Water Industries to determine whether theoretical implications can be practically observed. Finally, we segment the German energy market into three segments and analyze several case studies on specific businesses within each.

3. Energy sector mapping: hypothesis and methodology

This paper explores a key question, 25 years after Germany's energy liberalization and the introduction of various policies supporting renewable energy integration: have these structural changes increased the

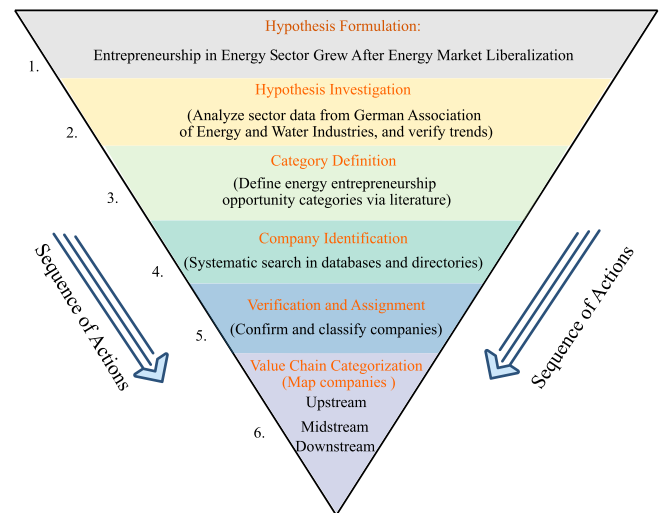


Fig. 2. A stepwise data-driven methodology for identifying entrepreneurial opportunities in the post-liberalization energy sector.

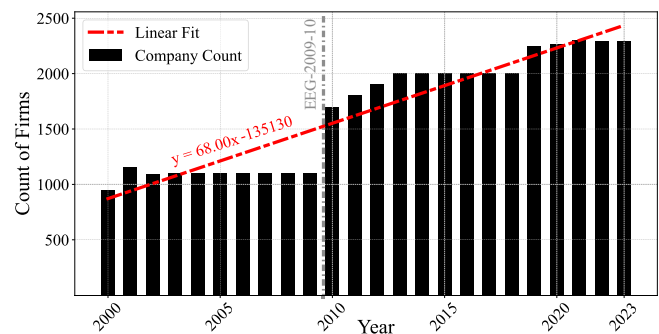


Fig. 3. Number of companies in the German energy market (2000–2023) (BDEW, 1998).

number of market participants? Understanding this is essential to evaluate how market reforms and renewable energy policies have influenced business growth and competition. If so, the paper further investigates which specific market categories have seen new business opportunities emerge. Fig. 2 outlines the research methodology used for this analysis.

3.1. Hypothesis formulation

The foundational hypothesis of this research is that the liberalization of the energy market and the supportive framework of the EEG stimulated growth in entrepreneurship within the energy ecosystem. This hypothesis posits a direct positive relationship between these policy shifts and the emergence of new ventures and business opportunities.

3.2. Initial hypothesis investigation

To investigate these hypotheses, we analyzed data from the German Association of Energy and Water Industries (Bundesverband der Energie- und Wasserwirtschaft, BDEW), spanning from 1998 to 2022. BDEW is the largest and most significant industry organization in the German energy sector, making its dataset both comprehensive and reliable. Fig. 3 illustrates statistics on the absolute number of active energy businesses in Germany, while Table A1 provides detailed counts across sub-sectors.

Changes in market structure and data collection policies have affected the consistency of yearly firm counts, particularly between 2009 and 2011. Despite this, a clear upward trend in the number of energy market firms in Germany is evident from 2000 to

2022 (see Fig. 3). A marked increase from 2010 onwards is closely linked to the EEG 2009 and the 2010 PV Novelle (German Federal Government, 2010), which improved investment security by guaranteeing feed-in tariffs and priority grid access for renewable energy producers. These policies significantly lowered entry barriers and encouraged the entry of new participants, including energy cooperatives, small and medium-sized enterprises, and individual prosumers, ultimately diversifying and expanding the number of active firms in the sector.

This underscores the theoretical implications of energy market liberalization theory. As previously mentioned, the dataset contains information on electricity, gas, and district heating. However, for the sake of a focused analysis, we will constrain our case studies to the electricity market (BDEW, 1998).

Next, we aimed to find specifically the companies operating within Germany's electricity sector, particularly those established or significantly adapting their business models after the liberalization of the energy market, with the aim of identifying the categories and subcategories that presented business opportunities.

3.3. Definition of categories and subcategories

Based on an extensive review of technical research papers, market reports, and the insights of subject matter experts, key areas where entrepreneurial opportunities are emerging have been identified. The primary drivers of these opportunities stem from technical challenges arising due to uncertainties introduced by market liberalization and the presence of multiple players. Additionally, the increasing penetration of renewable energy sources has resulted in issues such as reduced short-circuit power, decreased grid inertia, and complex interactions among control systems (Pant et al., 2024, 2025). These technical challenges have, in turn, led to a wide range of new business opportunities. To offer a clear overview, Table 2 organizes these opportunities into distinct categories and subcategories.

Table 2
Categorization of business opportunities in electricity sector.

Category	Subcategories
Renewable Energy	Solar/PV Hydroelectric Waste-to-Energy Portfolio Management
Energy Storage	Battery Storage (Stationary and Portable) Second-life Batteries
Energy Services	Virtual Power Plants (VPP)
Climate Tech	Energy Consulting and Management Services Carbon Capture, Utilization and Storage (CCUS) Sustainable Materials and Technologies Carbon and Sustainability Management Platforms
Smartgrid and Digitalization	Smart Meters and Energy Data Platforms AI and Analytics for Energy
Project Development	Renewable Energy Project Development
Grid Services	Grid Analysis and Optimization Software Development/Services Grid Automation and Control
E-Mobility	EV Charging Solutions and Infrastructure Hydrogen Mobility Solutions
Fusion Energy	Fusion Technology
Financing and Investment	Energy Project Financing and Investment Platforms
	Wind Energy Geothermal Renewable Energy Services and Integration
	Hydrogen Production and Storage
	Flywheel Storage Demand Response and Flexibility
	Energy as a Service (EaaS)
	Energy Efficiency and Savings
	Sustainable Construction and Buildings Carbon Trading
	Home Energy Management Systems (HEMS)
	IoT and Cloud Platforms for Energy Project Financing Grid Operation and Services (DSO/TSO related) Grid Stability and Ancillary Services Electric and Solar Vehicles
	Sustainable Mobility Platforms
	Agri PV Biomass/Bioenergy EPCs
	Power-to-X and E-Fuels
	Emerging Storage Technologies Energy Trading and Optimization Platforms Infrastructure Inspection and Maintenance Environmental Monitoring and Prediction Circular Economy Solutions
	Building Energy Management Systems (BEMS) Blockchain Applications for Energy
	Grid Infrastructure and Maintenance Solutions
	Battery Swapping Solutions

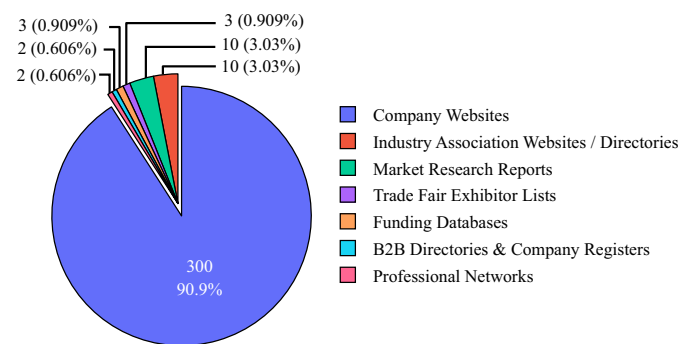


Fig. 4. Sources consulted for compiling list of companies.

3.4. Systematic identification of companies

A targeted search strategy was implemented using authoritative German and European sources to compile an initial list of potential companies. These included official company registers (e.g., *Unternehmensregister*), B2B directories (e.g., *wlw.de*), national industry association member lists (e.g., *BSW-Solar*, *BVES*, *BWE*, *VDMA*, *ZVEI*), market research reports (e.g., from *Fraunhofer Institutes*, *Agora Energiewende*, *dena*), trade fair exhibitor lists (e.g., *The Smarter E Europe*, *E-world*, *Hannover Messe*), professional networks (*LinkedIn*, *XING*), and public funding project databases (e.g., *Förderkatalog des Bundes*, *CORDIS*, *LIFE Programme*). The summary of consulted sources is presented in Fig. 4.

3.5. Verification and classification

The companies identified through these sources underwent a strict validation procedure, involving the following steps:

- Confirming an operational presence in Germany, primarily through official company websites and their “*Impressum*” (Legal Notice).

- Analyzing product descriptions, service offerings, and project portfolios to ensure clear alignment with one or more of the predefined entrepreneurial subcategories.
- Assigning each verified company to the most relevant primary subcategory based on its principal activities and focus within the German market.

This comprehensive process resulted in a curated list of over 240 companies, though it represents only a subset due to search limitations and keyword selection. This classification is visually summarized in Fig. A1. A comprehensive list of these ventures, along with the keywords used for identification and classification, is available in the Git repository (Prashantgrid, n.d.).

3.6. Value chain categorization

The categorization in the Table 2 is based on major technological fields and service areas formulated with the aim of identifying the ventures operating in those focus areas. However, one category or subcategory can span across multiple technical levels and scales, each bringing unique challenges and opportunities. For instance, in the residential storage segment (low-voltage systems), companies like **Sonnen GmbH** primarily focus on providing battery systems for individual households, operating at voltages typically between 230 V and 400 V. These solutions are optimized for self-consumption, backup power, and participation in residential-level virtual power plants. These systems comply with standards like VDE-AR-N 4105 (VDE, 2018), in Germany, and often leverage incentive schemes such as the KfW Residential Energy Efficiency Program (KfW Bankengruppe, 2024).

In contrast, large-scale storage providers such as **Fluence Energy** and **Wärtsilä Energy** specialize in high-voltage installations ranging from 10 kV to 400 kV, targeting grid stabilization, frequency regulation, and participation in balancing markets (Pant et al., 2024). These systems adhere to stricter technical standards such as VDE-AR-N 4100 (VDE, 2019), and EN 50549, (CENELEC, 2019) and must navigate complex national regulations of EnWG and EEG, particularly regarding network access, remuneration mechanisms, and ancillary service provision.

This distinction clearly demonstrates that even within the energy storage category, the business models, investment frameworks, technical designs, and regulatory compliance pathways diverge significantly depending on the system scale and voltage level. Thus, a structured categorization based on upstream, midstream, and downstream activities becomes essential to accurately map entrepreneurial opportunities and market entry strategies across the energy sector.

Upstream refers to the generation of electricity. Traditionally, this used to be industrial and small-scale auto producers generating electricity primarily for their own use (Lenz et al., 2019). In recent decades, the push for more sustainable energy systems has transformed the generation sector. Conventional power plants are increasingly being replaced by decentralized renewable energy sources, such as wind and solar (Clearingstelle EEG/KWKG, 2023). As a result, the generation space has become more dynamic, with a growing emphasis on sustainability and green energy.

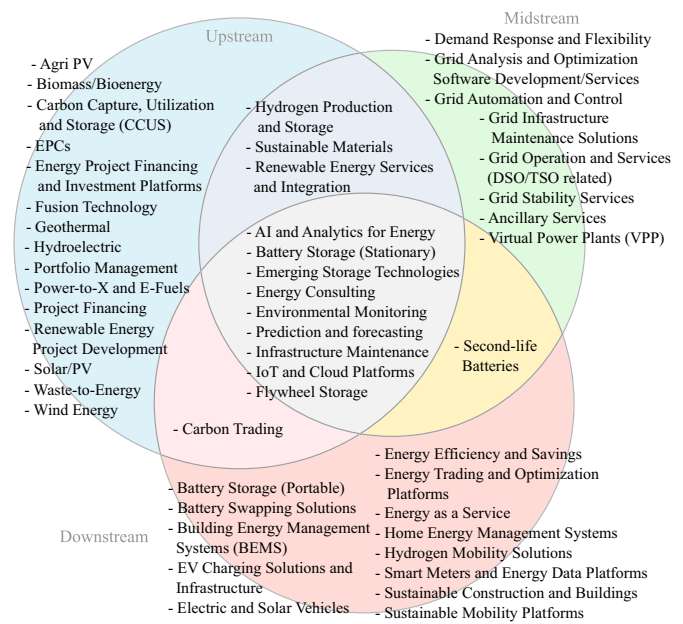


Fig. 5. Distribution of business categories within the upstream, midstream, and downstream segments of the electricity sector.

The **midstream** involves the transmission and distribution of electricity from generation points to consumers. The increasing integration of distributed energy resources is transforming the operation of power grids (Pérez-Arriaga and Bharatkumar, 2016). This introduces new challenges and complexities, requiring adaptations in grid management to maintain stability and efficiency (Jacobson, 2021). As the demand for ancillary services grows, technologies such as energy storage systems play a critical role in enhancing grid flexibility (Pant et al., 2024).

Operating at the low-voltage consumer end, the **downstream** energy segment involves companies delivering electricity and associated services directly to end-users. Here, growing competition enhances consumer choice and impacts pricing (Lenz et al., 2019), prompting firms to offer innovative solutions like energy management tools (Al-Ali et al., 2017). These tools can improve grid stability and boost self-consumption for lower electricity bills, in a market increasingly shaped by influential prosumers (Kotilainen et al., 2018).

Table 3 outlines the traditional definitions of upstream, midstream, and downstream segments, their evolution in response to emerging market trends, and examples of energy ventures within each category. Fig. 5 complements this by visually mapping business opportunities according to the value chain positioning of these ventures.

This framework allows us to map what opportunities emerged and where they fit within the value chain. Yet classification alone cannot answer a deeper question central to our causal argument:

Table 3

Market segments and key developments post-liberalization (Lenz et al., 2019; Pant et al., 2024).

Segment	Description	Key Changes Post-Liberalization	Technologies and Innovations	Example Energy Ventures
Upstream	Electricity generation	Emergence of independent power producers, Rise of renewables (wind, solar), More competition in generation	Renewable energy systems (wind, solar), Energy storage	Eon, Enerparc, Enel Green Power, Nordex, First Solar
Midstream	Transmission and distribution of electricity (including balancing)	Increased complexity due to renewable fluctuations, Focus on grid stability and energy security	Smart grids, Large-scale storage solutions, Ancillary services, Arbitrage trading, software solutions	Entrix, Voltfang, Entelios, RWE, TenneT
Downstream	Delivery of electricity to end users (consumers)	More consumer choice, Growth of prosumers, Emergence of innovative energy management solutions	Smart meters, Energy management tools, Smart homes	GridX, Enpal, Naturstrom, Sonnen, Tiko, Lumenaza

“Are these ventures genuinely products of liberalized market competition, or would they have appeared regardless as inevitable technical responses to renewable integration?”

3.7. System-driven vs. liberalization-dependent innovations

To clarify the causal drivers of new ventures in the German energy sector, our analysis is based on the Multi-Level Perspective (MLP) on socio-technical transitions (Geels, 2002). The MLP distinguishes between landscape pressures, regime dynamics, and niche processes, and shows how innovations can be (i) responses to physical system pressures embedded in landscape/regime dynamics, or (ii) products of niche formation and institutional openings that depend on market and regulatory design. Using this lens helps us distinguish between technical requirements and the institutional factors that determine who develops innovations and which business models are able to scale.

The MLP is particularly well-suited for this investigation because it mirrors the structure of our analysis: we examine how market liberalization and renewable energy policies at the landscape (macro) level shape the German electricity sector at the regime (meso) level, and how these changes create entrepreneurial opportunities and enable new venture emergence at the niche (micro) level. In other words, our study explicitly spans the three levels identified by the MLP.

System-driven innovations: Some technologies and services emerge primarily as technical responses to changing system conditions (e.g., high shares of variable renewable energy (VRE) increasing requirements for frequency control, reserves, storage, and forecasting). Technical assessments and empirical studies indicate that these physical needs stem from the evolving techno-physical landscape and regime interactions and would arise under alternative market structures (e.g., vertically integrated incumbents) as well as under liberalized markets (Rechsteiner, 2021; International Energy Agency, 2024; Bundesnetzagentur, 2024; Joskow, 2008; International Energy Agency, 2020). However, while the technical requirement is system-driven, the organization of provision (incumbent vs. third-party provider), the business model, and the pace and diversity of technological development can be strongly influenced by market design and regulatory rules.

Liberalization-dependent innovations: Other innovations rely on institutional features produced by liberalization, open retail markets, third-party access, customer choice, and marketplace participation rules. Typical examples are residential BESS operated as virtual power plants (e.g., Sonnen), distributed aggregation platforms (e.g., Next Kraftwerke), and subscription-based solar offerings (e.g., Enpal). These business models require regulatory recognition of third-party market access, aggregation and settlement rules, and rights to customer data and are therefore unlikely to scale (or to take the same form) under a monopolistic procurement model (Rechsteiner, 2021; Burger et al., 2019; Sousa et al., 2019).

A spectrum view: Following the MLP, we treat the distinction as a spectrum rather than a strict binary. In many cases, the underlying need is system-driven, while the commercial innovation (actors, bundling, financing) depends on liberalization. Our empirical classification, therefore, reports the primary driver (system vs. liberalization) but notes where an innovation combines both drivers (e.g., utility restructuring where VRE pressures create technical needs and liberalization permits strategic repositioning). Table 4 summarizes selected ventures and the rationale for their classification; the table's footnote explains the counterfactual test used to assign the primary driver.

New ventures are empirically concentrated in downstream segments (retail, aggregation, prosumer services, and distributed installation), which reflects the enabling effect of market-opening measures on customer-facing business models (Rechsteiner, 2021). Open retail markets, customer switching options, and freedom to design differentiated tariffs and service bundles facilitated models such as community solar and self-consumption; comparative evidence indicates these models

are more likely to emerge and scale in liberalized settings than under monopolistic procurement (Rechsteiner, 2021; International Energy Agency, 2022; Ruester et al., 2014).

By contrast, many midstream and upstream innovations (e.g., grid-scale storage and frequency-control technologies) primarily respond to techno-physical needs created by VRE integration and would very likely have been required under alternative institutional designs. (International Energy Agency, 2024; Bundesnetzagentur, 2024; Joskow, 2008) Liberalization shapes who provides such solutions and in what commercial form, but the underlying need is system-driven (Pant et al., 2024; Brunekreeft, 2000). What is distinctive about liberalization is its opening of the *demand side*: active participation of households and small businesses as prosumers, aggregators, and flexible consumers depends on open retail markets and customer choice. This downstream engagement marks the clearest break from a historically supply-driven electricity system and explains why the concentration of entrepreneurial activity is greatest in the downstream segment (International Energy Agency, 2022; Marş et al., 2020; Sareen and Haarstad, 2022; GridX, 2024h).

4. Cause-effect analysis: German energy policies and new business opportunities

The increasing penetration of intermittent renewable resources, combined with the involvement of multiple players in the electricity wholesale and generation markets, has given rise to new technical challenges. These challenges have driven the development of technical solutions, which in turn have created new business opportunities. However, while these technologies enable major changes and improvements, they do not initiate them independently. Instead, technological advancements primarily respond to needs arising from new regulations and economic developments (Kappner et al., 2023). To better understand the cause-effect relationship between policy interventions and business growth in the German electricity market, we present a timeline overview of key energy policies and their impacts on business development (see Fig. 6), serving as a reference for tracing the evolution of market opportunities and business dynamics.

4.1. EEG 2010 amendment: reinforcement of feed-in tariffs and market integration

The 2010 EEG amendment (based on the law passed in 2009 and effective from January 2010) reinforced stable, long-term feed-in tariffs, providing 20 years of above-market remuneration for renewable electricity. This significantly de-risked investments in technologies such as solar PV, wind, and biomass, making projects more attractive to investors and financiers (Deutscher Bundestag, 2010). As a result, it catalyzed the growth of project developers, EPC (Engineering, Procurement, and Construction) firms, and asset managers (Rövekamp et al., 2021).

Companies such as **Soventix GmbH**, founded in 2010, rapidly expanded their solar and wind development portfolios by leveraging the investment certainty provided by feed in tariffs. These tariffs guaranteed fixed payments for renewable generation, lowering financial risks for new developers. Meanwhile, incumbent players such as **BayWa r.e.**, operating as an independent power producer (IPP), capitalized on FiT-driven revenue stability and later complemented this with long-term power purchase agreements (PPAs) to diversify and secure market-based income streams, enabling them to systematically scale their asset management operations and renewable project pipelines (PV Tech, 2021).

At the same time, the EEG enabled the emergence of new business models for independent renewable energy suppliers (Xia-Bauer et al., 2022). **Polarstern Energie GmbH**, founded in 2011, built its model around delivering 100 % certified green electricity and gas, targeting both environmental and social impact (Roka et al., 2013). Its viability stemmed from three key features of the EEG: (i) guaranteed feed-in

Table 4
Classification of German energy ventures by primary innovation driver.

Case Study	Rationale
System-driven	
<i>Enerparc</i> (Utility-scale PV development)	Growth is linked mainly to FiTs and PV cost reductions. The underlying projects would occur under monopoly procurement models.
<i>Grid service providers</i> (Stabilization and reserve provision)	Provision of frequency and reserve services is required by VRE variability (system-driven). The technology development itself does not rely on liberalization; only the involvement of third parties is market-dependent.
Liberalization-dependent	
<i>Sonnen</i> (Residential BESS & VPP)	Business model relies on access to retail customers, aggregation rules, and participation in balancing markets, which are enabled by liberalization.
<i>Next Kraftwerke</i> (VPP aggregating distributed units)	While balancing needs are system-driven, the third-party aggregator model requires liberalized market access.
<i>Enpal</i> (Residential solar subscription)	The innovative financing model depends on an open retail market and consumer choice.
<i>Demand-response aggregators</i> (Pooling flexible loads)	Requires regulatory recognition for third parties, market access, and rights to customer data.
Mixed (System + Liberalization)	
<i>E.ON</i> (Utility restructuring)	Transformation was necessitated by VRE integration (system), but the ability to reposition into competitive retail and distributed services was facilitated by liberalization.

Note: Classification reflects the **primary driver** using the following counterfactual rule: would the innovation (or its business model) plausibly arise and scale under a vertically integrated (monopolistic) system? “Primarily system-driven” indicates the underlying technical need would likely exist independent of market design, though the organization and provider of that service may differ. “Primarily liberalization-dependent” indicates the business model, third-party market access, or customer-facing actor configuration likely requires institutional openings produced by liberalization (e.g., open retail markets, aggregation rules, rights to customer data). “Mixed” denotes cases where both system pressures and liberalization-enabled opportunities contributed substantially.

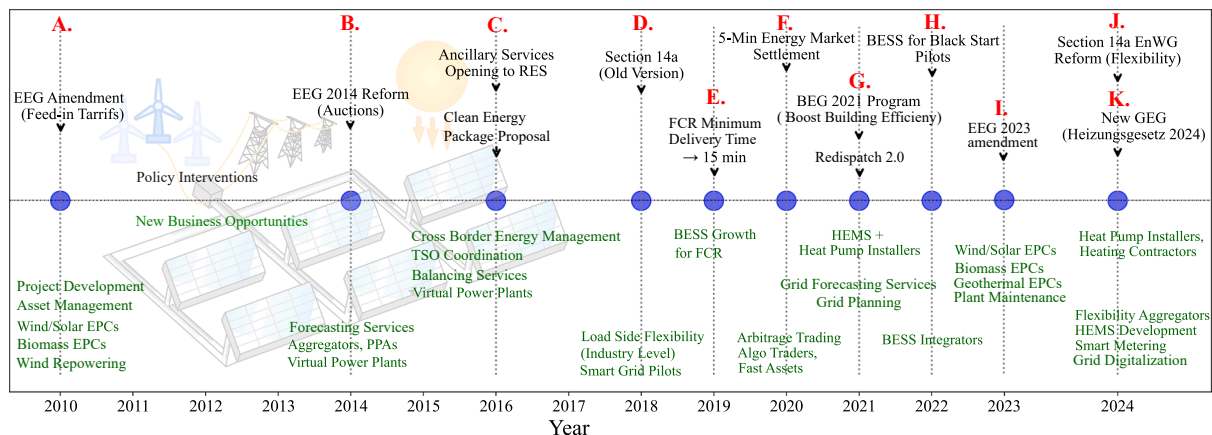


Fig. 6. Policy interventions driving innovation: a timeline of emerging business opportunities in the energy sector (2010–2024).

tariffs and priority grid access ensured a reliable supply of renewable power; (ii) market liberalization and unbundling allowed new, small-scale actors to compete without owning generation assets; and (iii) early incentives for direct marketing of renewable electricity, enabled transparent sourcing from specific green plants, supporting Polarstern's differentiation as a provider of verifiably sustainable energy.

4.2. EEG 2014 reform (Auctions)

EEG 2014 amendments marked the beginning of a transition from administratively set feed-in tariffs to competitive auctions for determining financial support for renewable electricity generation. This shift began with pilot auctions for ground-mounted solar PV systems above 1 MW, first implemented in 2015 (Wigand et al., 2016). The auction mechanism aimed to enhance cost-efficiency, market integration, and transparency in renewable energy expansion. Building on the pilot's outcomes, the

EEG 2017 reform institutionalized auctions as the primary allocation tool for new capacity across key technologies, including onshore wind, solar PV, and biomass (Agora Energiewende, 2016).

The evolution of EEG catalyzed the emergence of innovative business models. A prominent example is aggregators such as **Next Kraftwerke** (FOCUS Business, 2020), which operate virtual power plants, which are advanced software-based platforms that coordinate a wide range of decentralized energy assets. These systems enable distributed generators to actively participate in various energy markets. This includes bidding aggregated capacity into balancing power markets to support grid stability, optimizing electricity sales in short-term wholesale markets, and complying with the mandatory Direct Marketing requirements for renewable energy under the EEG framework (LoSection Ner et al., 2017).

In 2014, Next Kraftwerke reported revenues of approximately €180 million and operated a VPP with 1500 MW of capacity (Menrad, 2015). By 2018, the company had grown substantially, managing a network

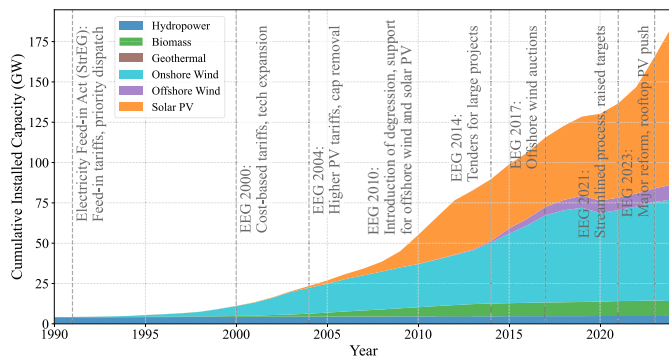


Fig. 7. Germany's renewable capacity (1990–2024) with key EEG interventions, data retrieved from Burger (2024).

of 8100 decentralized plants with a combined capacity of 7100 MW, reflecting a rapid expansion in both operational scale and market influence within just four years (FOCUS Business, 2020).

Simultaneously, other companies like **Enpal** focused their strategy on the residential rooftop solar segment. This market, involving small systems on individual homes, remained largely exempt from the mandatory auction mechanism applicable to larger projects. Enpal pioneered innovative solar subscription offerings (Enpal GmbH, 2024). These are typically long-term leasing or rental models where homeowners avoid the high upfront purchase cost of installing solar panels (on average $\approx$ €15,000). Instead, they pay Enpal a fixed monthly fee over many years. In return, the homeowner uses the solar electricity generated, saving money on their utility bills, ideally more than the subscription costs. Crucially, this model allowed Enpal to rapidly deploy residential solar by targeting a segment unaffected by the main auction rules and offering an accessible financial package, thus effectively bypassing the primary auction mechanism and its associated competitive pressures.

While this section has focused on the pivotal EEG 2014 reform and its subsequent impact on renewable business models, it is important to recognize that the EEG has been amended multiple times since its original enactment in 2000. Each amendment, whether in 2000, 2004, 2009, 2012, 2017, 2021, or 2023, introduced targeted adjustments to tariff structures, market mechanisms, and investment incentives. These policy refinements have collectively shaped Germany's renewable energy landscape, steadily expanding installed capacity and encouraging significant private investment across technologies. Fig. 7 illustrates the cumulative installed renewable electricity capacity in Germany from 1990 to 2024, highlighting key EEG interventions that have catalyzed this growth.

4.3. Ancillary services opening to renewable energy sources (2015–2016)

From 2015 onward, regulatory adaptations (especially prequalification changes by TSOs and BNetzA) allowed renewable energy sources and distributed assets to provide ancillary services, including FCR and secondary reserves (Bundesnetzagentur, 2015). This broke the historical dominance of fossil-based synchronous generators. Aggregation models became commercially viable (Birk et al., 2021; Mao et al., 2020).

Next **Kraftwerke** expanded its VPP platform to monetize flexibility from decentralized producers, while **Sonnen GmbH** aggregated residential batteries into its “sonnenCommunity” to participate in balancing markets such as frequency containment reserve (FCR), providing grid services previously inaccessible to small-scale assets (TransnetBW GmbH, 2024).

4.4. Section 14a EnWG (2017)

In 2017, Section 14a of the EnWG permitted DSOs to enter into voluntary agreements with end consumers, allowing DSOs to remotely control certain controllable loads such as heat pumps, electric vehicle

chargers, and night storage heaters, to manage local grid congestion. In exchange, consumers benefited from reduced network tariffs. This approach aimed to prevent localized overloading in the low-voltage grid by utilizing consumer flexibility. While the application of this provision was initially limited and voluntary, it marked a significant step toward integrating demand-side flexibility into grid management strategies.

The regulatory support for controllable loads under Section 14a EnWG encouraged the development of smart load control technologies and demand response (DR) business models. For example, **Entelios** became Germany's first demand response aggregator, pooling electrical loads from commercial and industrial energy users (Stede et al., 2020). The company developed VPPs and marketed aggregated capacity to grid operators, facilitating demand-side flexibility in the energy market.

Another venture called **GreenCom Networks**, mainly focuses on providing residential energy management platforms that connect home energy devices, such as PV systems, batteries, and heat pumps, to VPPs. Their software solutions enable distributed residential assets to participate in demand response programs and grid services. GreenCom collaborated with utilities and device manufacturers across Germany and Europe (New Power, 2019), before being acquired by Enphase Energy, Inc. in 2022 (AI Research Tool, 2022). The acquisition aimed to expand Enphase Energy's comprehensive home energy management solutions and flexibility services, particularly within the European market.

4.5. FCR minimum delivery time reduction (2018)

In 2018, Germany, following European Network of Transmission System Operators for Electricity (ENTSO-E) guidelines, shortened the minimum delivery period for FCR services from 30 to 15 min (ENTSO-E, 2017). This change favored BESS systems, as they can respond faster and supply energy for a short duration of 15 min (Englberger et al., 2020; Zou et al., 2025).

Sonnen capitalized on this opportunity by aggregating residential battery systems into virtual FCR pools, enabling prosumers to participate in ancillary service markets and earn additional income (E. S. News, 2023). **Tesvolt** expanded its industrial-grade storage solutions tailored for FCR provision, targeting commercial and industrial sectors (Tesvolt AG, 2025).

At the same time, a new wave of BESS innovators emerged: **STABL Energy GmbH** developed modular second-life battery systems optimized for rapid grid services; **Fenecon GmbH** introduced scalable energy management solutions and second-life battery integrations; and **ecostor GmbH** provided high-performance stationary storage systems designed specifically for flexibility and FCR markets (ecostor, 2025). These developments expanded the pool of distributed assets capable of meeting the new 15-minute delivery requirements, enhancing system reliability and accelerating the integration of decentralized storage into grid operations.

4.6. 5-minute energy market settlement (2019)

Following European balancing market harmonization (Electricity Balancing Guideline, EBGL), Germany implemented a 5-minute settlement period in 2019 (European Commission, 2017). Shortening settlement intervals increased volatility and amplified the value of fast-reacting, flexible assets in wholesale and balancing markets. This created opportunities for arbitrage, as flexible assets could now capitalize on intraday price differences between energy markets. Algorithmic and real-time trading platforms became essential for exploiting these price spreads.

Entrix Energy developed real-time asset optimization algorithms, while **Next Kraftwerke** enhanced its VPP platform with short-term forecasting and algorithmic bidding tools to maximize revenue opportunities for decentralized producers (Mao et al., 2020; LoSection Ner et al., 2017). Additionally, companies like **Tiko Energy Solutions**, **RWE**

Supply and Trading, and **Statkraft** have successfully leveraged arbitrage strategies to optimize energy trading and increase profitability within the volatile 5-minute settlement framework.

Furthermore, battery analytics companies like **TWAICE Technologies** (TWAICE, 2025), based in Munich, offer advanced predictive software that models battery aging, state of health, and performance under varying operational conditions. Their solutions allow storage operators and electric vehicle fleet managers to optimize dispatch strategies, extend asset lifespans, and reduce degradation-related costs. This enables participation in ancillary service markets (Pant et al., 2024), and price arbitrage in the 5-minute intraday market. Similarly, **ACCURE Battery Intelligence** (Aachen) specializes in battery monitoring and risk management software for large-scale operators.

4.7. BEG 2021 program and redispatch 2.0 (2021)

In 2021, Germany introduced the BEG (Federal Subsidy Program for Efficient Buildings) to promote energy efficiency in the residential and commercial sectors. The program provides financial incentives for upgrading building energy systems, notably supporting the installation of heat pumps and solar thermal technologies to reduce carbon emissions and improve energy performance (ZSW, 2024). A key objective is to electrify 50 % of the heating load by 2030, primarily via heat pump deployment, as part of Germany's strategy to decarbonize the building sector and achieve carbon neutrality by 2045. Subsidies include a 30 % base grant for replacing fossil-fuel heating systems, an additional 5 % bonus for systems using natural refrigerants or renewable heat sources, and up to a 20 % bonus for replacing oil, coal, or gas systems older than 20 years. Income-based incentives offer up to an additional 30 % for lower-income households. These measures aim to accelerate market adoption of energy-efficient technologies and support national climate goals (BMWK, n.d.).

At the same time, the implementation of Redispatch 2.0 expanded the grid management responsibilities of TSOs. Under the new regulations, all generation units greater than 100 kW, including renewable energy sources, were now required to be part of the redispatch process (Hirth et al., 2021). Redispatch 2.0 mandated that these generation units could be curtailed or adjusted to manage grid congestion and ensure stability in the electricity system. This was a crucial step in integrating variable renewable energy sources such as wind and solar into the grid more effectively, while maintaining grid reliability and minimizing curtailment losses. These two complementary policies created dual business growth pathways:

4.7.1. Energy retrofit services and heat pumps

The heat pump installations are seeing an uptrend in Germany and are regarded as business hot spots at the moment (Germany Trade & Invest Gesellschaft für Außenwirtschaft und Standortmarketing mbH, 2023; Golly et al., 2022). Also, companies such as **Thermondo** have capitalized on the subsidies for energy efficiency improvements, offering turnkey services for residential and commercial buildings to enhance their energy performance. Thermondo specializes in the installation of heat pumps, energy-efficient heating systems, and insulation upgrades. The subsidies under the BEG program directly target segments that would benefit from these improvements, such as homeowners and property developers (Thermondo GmbH, 2025). The policies have also created a thriving market for start-ups focusing on building analytics (**Lumoview**, **QEA Tech**) or specialized financing advisors (**PuR GmbH**) which find opportunities by supporting the planning and execution of subsidized energy transition projects. These companies demonstrate how effectively integrating policy knowledge and subsidy access into their business models can unlock significant market potential by reducing investment barriers for consumers.

4.7.2. Decentralized energy resource (DER) management

The rise of Redispatch 2.0 created new opportunities for technology providers like **GreenCom Networks**, which developed advanced DER management platforms to support the integration of decentralized energy generation (e.g., solar PV, wind) and flexible loads into grid operations. These platforms enabled DERs to participate in grid balancing and congestion management by responding to grid signals for generation adjustments or load shifting. While **GreenCom Networks** was an important early innovator in residential DER management solutions following the implementation of Redispatch 2.0, larger aggregators such as **Next Kraftwerke** (Next Kraftwerke, 2023), **gridX GmbH**, and **Venios GmbH** dominated the broader market for commercial and industrial asset integration (IWR Pressedienst, 2021).

4.8. BESS for black start pilots (2022)

TSOs initiated pilot projects in 2022 to test the viability of BESS for black start capabilities, traditionally reserved for diesel generators (Amprion GmbH, 2022; Klein et al., 2023). Regulatory approval for grid-forming inverter systems has created a novel application for utility-scale and industrial battery projects.

Companies such as **Fluence Energy** supplied large-scale storage solutions for pilot tests, while **Tesvolt** (TESVOLT AG, 2025) and **STABL GmbH** adapted their inverters and control algorithms for black start and microgrid applications, expanding the addressable market for resilient energy infrastructure (Sungrow Power Supply Co., Ltd).

4.9. EEG 2023 amendment

The 2023 revision of EEG, known as the “Easter Package,” significantly accelerated the country's energy transition by setting a binding target of at least 80 % renewable electricity by 2030. Financially, a fundamental shift occurred through the complete abolition of the EEG levy as of January 1, 2023 (following its interim reduction to zero in July 2022) (Bundesregierung, 2023). Renewable energy funding responsibilities were transferred to the federal Energy and Climate Fund (EKF), with the government estimating €6.6 billion in costs for the second half of 2022 alone. The amendment further expanded auction volumes for wind and solar energy and introduced simplified regulations for citizen energy projects.

The EEG 2023 also strategically promoted diversification of the renewable energy mix by enhancing support for geothermal energy, recognized for its baseload generation capabilities in contrast to intermittent wind and solar sources. Geothermal projects received a feed-in tariff of 25.20 ct/kWh (subject to a 0.5 % annual degression starting in 2024) and regulatory easing through potential exemption from the Federal Mining Act (Lietz, 2023).

While established players such as **juwi AG** and **BayWa r.e.** remain dominant in wind and solar development, these policy changes have catalyzed new market entrants and expansions. For example, **Eavor GmbH**, the German subsidiary of Canadian geothermal company Eavor Technologies, established operations in Germany to capitalize on the enhanced geothermal support (Eavor Technologies Inc, 2025). In the solar sector, companies like **Enpal GmbH** expanded their service offerings in response to growing demand driven by EEG reforms (Radke, 2024), offering lease-based residential solar systems at scale. Similarly, **1Komma5° GmbH**, founded in 2021 but significantly scaling operations post-2023 EEG amendment (Calilao, 2024), has rapidly built a business around integrated solar, storage, and energy management systems targeting households and small businesses.

4.10. Section 14a EnWG reform (2024) (Flexibility)

The reformed Section 14a of the EnWG (2024) mandates the technical readiness for the active control of flexible loads, including those up to 4.2 kW, enabling DSOs to dynamically manage energy consumption

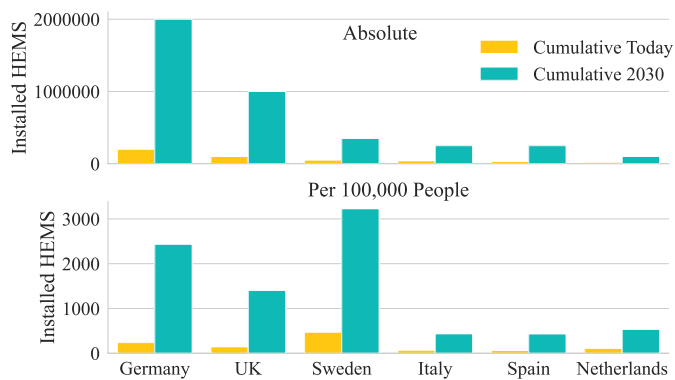


Fig. 8. Expected growth in installed HEMS, data retrieved from GridX (2024j).

during critical grid situations (Bundesnetzagentur, 2024). This regulatory change creates opportunities for businesses to develop innovative solutions to support grid stability, such as smart metering systems and HEMS.

For example, **Tibber** capitalized on this by expanding its smart metering offerings with flexible tariffs, while **GridX** developed integrated HEMS to optimize energy use based on real-time grid signals (GridX, 2024j). These companies emerged and grew due to the demand generated by the policy's focus on grid stability and flexible energy solutions.

Fig. 8 illustrates the projected growth of HEMS in selected European countries by 2030. Germany is expected to lead in absolute installations (approximately 2 million), primarily due to its larger population. However, when adjusted for population size, Sweden exhibits the highest projected penetration rate per 100,000 inhabitants. As these technologically advanced nations exhibit such varied adoption forecasts, it strongly indicates that market development hinges on distinct national contexts rather than uniform technological availability. Key differentiating factors likely include specific energy policies (e.g., feed-in tariffs vs. net metering), financial incentive structures, the maturity of related markets like rooftop solar PV or electric vehicles, and diverse consumer attitudes (Berger et al., 2022), environmental concerns, and energy independence goals. Consequently, effective HEMS business models must be localized to address the distinct realities of each national market.

On the utility and industrial side, network operators and aggregators developed new flexibility management systems and demand response platforms to comply with and benefit from Section 14a EnWG. For example, **EnBW Utility Services**, for instance, adopted **Robotron's** CLS (Controllable Local Systems) management software to meet Section 14a requirements. The platform connects approximately 150,000 smart-meter endpoints (including wallboxes and heat pumps) into a central control system, enabling DSOs to manage local congestion while facilitating immediate connections and applying tariff discounts (Robotron Datenbank-Software GmbH, 2023). Likewise, demand-response aggregators expanded in Germany: the Dutch firm **Sympower** raised €21.3 million in 2022 to deploy its flexibility management platform across the EU. The company now aggregates more than 2 GW of industrial and commercial loads for balancing markets and grid services, highlighting investor confidence in regulatory frameworks such as Section 14a that monetize customer-side flexibility (Sympower, 2022).

The Section 14a amendment has also accelerated the development of VPPs. **Sonnen**, now a subsidiary of **Shell**, has networked over 25,000 home battery units, equating to roughly 250 MWh of distributed capacity, into what has become Germany's largest residential VPP by 2023, with plans to scale toward 1 GWh. These batteries are now designed to be remotely managed during Section 14a-related grid events to maintain voltage stability and defer infrastructure upgrades (E. S. News, 2023). Similarly, venture-backed battery developers like **Green**

Flexibility have emerged: in 2025, **Partners Group** committed €400 million to the company, which aims to build over €1 billion worth of battery energy storage projects in Germany. Executives cite Section 14a's incentives, in combination with the solar deployment surge, as critical enablers for this business model (E. S. News, 2025).

4.11. New Gebäudeenergiegesetz (GEG), or building energy act

Germany's building energy policy experienced a major transformation with the introduction of the GEG in 2020 (effective November 1, 2020) (GEG, n.d.). This legislation consolidated previous energy-saving laws and charted a clear path toward achieving a climate-neutral building stock by promoting energy efficiency and the use of renewable energies, thereby establishing a foundation for subsequent, more stringent regulations. While the GEG 2020 mainly unified and updated existing standards without imposing radical new obligations on heating systems, a major shift occurred with the GEG amendment of 2024 (commonly referred to as the "Heizungsgesetz"). This amendment, largely effective from January 1, 2024, introduced a new legal requirement: newly installed heating systems must source at least 65 % of their energy from renewable sources. The obligation is applied with staggered timelines, generally targeting adoption from 2026 or 2028, depending on municipal heat planning and building type. This marked a decisive move away from fossil-fuel-based heating toward a renewable energy supply (Deutsche Energie-Agentur, 2024).

This strong policy shift toward renewables had a profound impact on the market, spurring the creation of new businesses and the strategic evolution of existing ones, particularly within the heat pump sector (Verheyen and Graf, 2025; GridX, 2024j). A notable example is the establishment of **Qvantum Energietechnik GmbH**, established in June 2023, right as the GEG 2024 ("Heating Law") was being finalized, indicating the law's role as a driving factor (Qvantum, 2023). Concurrently, established companies like **Tyczka Energy**, traditionally focused on gas-based heating solutions, adapted by diversifying into renewable energy technologies, including heat pumps. Tyczka's earlier collaboration with **Viessmann**, initiated in 2021 in the context of the GEG 2020, to develop hybrid heating systems further illustrates how businesses proactively responded to the evolving regulatory environment. Together, these developments highlight how Germany's policy-driven shifts have stimulated both entrepreneurial activity and strategic adaptation, fostering innovation in sustainable and energy-efficient technologies.

This section has shown how energy market liberalization and supportive policy frameworks have driven the emergence of new ventures across Germany's electricity sector. From 2010 to 2024, regulatory changes catalyzed significant growth in upstream activities, notably in large-scale PV, onshore/offshore wind, and geothermal projects, domains traditionally dominated by established EPCs requiring substantial capital investment. In contrast, the midstream and downstream sectors offer compelling entrepreneurial prospects, characterized by lower entry barriers and abundant innovation opportunities in grid services, VPPs, demand aggregation, DER management, residential solar, heat pumps, and HEMS. To illustrate how these trends translate into concrete business outcomes, the next section presents detailed case studies of selected companies across the value chain, providing a clearer analysis of how policy interventions have shaped their development.

5. Case studies: upstream

The case of **E.ON** is a good example for understanding how regulatory changes and market liberalization directly influence the strategic transformation of incumbent energy companies. It offers valuable insights into how policy particularly the EU's internal energy market liberalization, Germany's nuclear phase-out, and strong support for renewable energy can act as a catalyst for corporate restructuring and the emergence of new spin-offs and business models. As such, this case

is central to analyzing the broader impact of energy policy on the formation of new energy ventures.

5.1. E.ON

E.ON SE was established in June 2000 as a result of the merger of two large industrial conglomerates, VEBA AG and VIAG AG. Both firms previously had significant operations in oil and gas exploration and production, as well as electricity generation and distribution. To consolidate their energy operations and become a dominant force in the European energy market, the merger was carried out in 2000, and E.ON was born (Chatterjee et al., 2023). Further acquisitions of Powergen and Ruhrgas followed throughout 2002 and 2003. At this point, E.ON's goal was to become the world's leading integrated power and gas company (E.ON SE, 2024).

E.ON's 2010 s were transformative, as the strategy for the E.ON group was changed to "Cleaner and Better Energy," under which the group aimed to grow outside of Europe and focus on reducing CO₂ emissions. In 2011, E.ON started the decommissioning and dismantling of its nuclear power plants. In 2014, it was announced that E.ON would no longer generate energy from fossil fuels, focusing instead on renewable energies, energy grids, and customer solutions. This strategy was implemented in 2016 when the nuclear and fossil fuel business was spun off into Uniper, which was then acquired in 2018 by Fortum. In 2020, E.ON acquired Innogy's grid and sales business (E.ON SE, 2024).

Today, E.ON employs 75,000 people, serves 47 million customers, and has an adjusted EBITDA of €9.4 billion (E.ON SE, 2024).

E.ON started as a vertically integrated supplier and now operates in midstream and downstream markets. This convoluted history and the multiple 'reinventions' that E.ON underwent can be directly traced back to three fundamental policy changes that significantly impacted E.ON's strategic decisions over the past 25 years.

The first was the EU Electricity and Gas Directives and the EnGW in 1998. The goal here was to liberalize the European and German energy industries and break up regional monopolies to encourage competition (Bundesnetzagentur, 2024). They mandated the unbundling of vertically integrated utilities, separating generation, transmission, and distribution. To remain competitive in the liberalized markets, VEBA AG and VIAG AG merged into E.ON in 2000 to leverage economies of scale and streamline operations. E.ON maintained ownership of assets across the value chain because they provided third-party access to their transmission networks and kept separate segments' financial accounts (Anchustegui, 2018).

The second significant strategy shift resulted from the EEG and the German government's decision to phase out nuclear power in 2011. The FiT was a core mechanism in the EEG and ensured that renewables were granted priority access to the grid. As a result, E.ON changed its investment strategy, shifting from fossil fuel-based and nuclear power generation to renewable energy development instead (E.ON, 2023). Finally, in 2016, the EU Third Energy Package was fully implemented, completing the EnGW of 1998 and enforcing strict unbundling and separation of energy supply and generation from transmission operations. To comply, E.ON spun off its fossil fuel and nuclear business into Uniper in 2016 (E.ON SE, 2018), focusing on transmission and customer solutions in Europe.

This culmination of policy pressures also spurred the landmark asset swap between E.ON and RWE, initiated in 2018 and largely completed by 2020, as both sought to specialize within the evolving regulatory landscape. In essence, E.ON acquired Innogy (RWE's former subsidiary), retaining Innogy's extensive electricity and gas networks and its customer solutions business, thereby cementing E.ON's strategic shift to a focused operator of energy infrastructure and customer-centric services. In return, RWE absorbed the renewable generation portfolios of both

E.ON (its existing significant renewable assets) and Innogy, transforming RWE into one of Europe's leading renewable energy producers. This complex realignment allowed each company to pursue a more defined strategic path consistent with market and policy trends (Rack, 2020).

Reflecting its new strategic focus on customers, networks, and energy solutions, E.ON launched a series of subsidiaries and initiatives that align closely with evolving European and national energy policies:

1. *E.ON Agile Accelerator* (2013): Initiated to foster energy startups and innovation in response to the EU's Horizon 2020 program, supporting sustainable technology development.
2. *E.ON Drive* (2016): Developed to expand electric vehicle charging infrastructure, aligning with the EU Alternative Fuels Infrastructure Directive (2014/94/EU), which aims to accelerate e-mobility adoption.
3. *E.ON Energy Infrastructure Solutions (EIS)* (2016): Created to deliver decentralized district heating, cooling, and energy networks, in line with the EU Energy Efficiency Directive (2012/27/EU).
4. *E.ON Home* (2019): Launched to provide smart home energy systems and residential energy optimization, driven by Germany's national energy efficiency strategies and EU directives encouraging digital and efficient energy consumption.
5. *E.ON One* (2022): A digital platform for grid and energy service optimization, aligning with the EU's Digitalization of Energy Action Plan, which supports smart grid development and data-driven energy management.

This sequence of ventures illustrates how policy not only forced legacy companies like E.ON to restructure but also opened up new entrepreneurial spaces within the energy sector. E.ON's evolution from a traditional utility to a multi-platform energy solutions provider reflects a broader trend in which liberalized markets and decarbonization policy frameworks enable the formation of new, future-oriented business models.

5.2. Enerparc AG

Enerparc AG is a German company founded in 2008, which focuses on the development, construction, and operation of large-scale solar power plants, as well as the marketing of the electricity they produce (Enerparc AG, 2024). The Hamburg-based company is considered a market leader in the German solar power field, having installed 4.5 GW of solar power worldwide, 3.5 GW of which it sells into the German energy market (Lux, 2024).

Enerparc's founding was heavily influenced by Germany's FiT system, introduced under the EEG. These tariffs reduced investment risks and enabled the rapid scaling of solar projects on both large and small scales. This enabled Enerparc to calculate clear ROI cases for its large-scale solar parks, reducing risks and increasing the number of financing options.

Fig. 9 shows the development of the FiT for large-scale solar parks, starting at approximately 50 cents/kWh in 2000, and decreasing as the production costs of solar decreased and the energy source became more competitive.

The establishment of the FiT for large and small-scale solar power plants was extremely effective across multiple European states. When compared to other policies like quotas, it can be seen that guaranteed FiTs are a successful instrument for promoting the build-up of renewable energy sources in both terms of market growth and economic efficiency for the state (Vienna University of Technology, 2004; Mario, 2005). As shown in Fig. 10, since its introduction, the German solar market has experienced rapid growth and has made space for players like Enerparc, who provide electricity for one million homes.

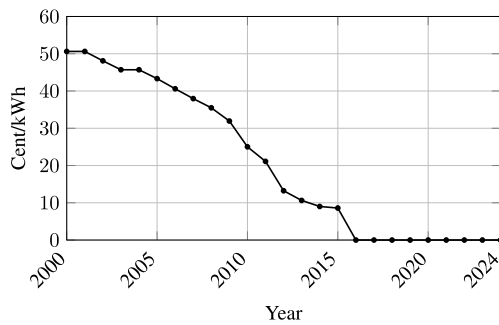


Fig. 9. Feed-in-Tariff for open space solar facilities (Deutscher Bundestag, 2022; SFV, 2012).

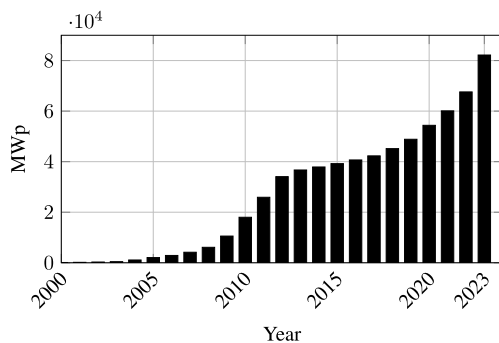


Fig. 10. Installed capacity of photovoltaic systems in Germany (2000–2023, in megawatt peak) (Umweltbundesamt, 2024).

6. Case studies: midstream

6.1. Entrix

Entrix is a German startup founded in 2022, specializing in AI-driven energy storage optimization and energy flexibility. Their software platform enables businesses owning BESS assets to manage their systems efficiently by controlling battery usage, for example, through intelligent use of energy price arbitrage throughout the day. The platform integrates the financial energy markets with battery hardware and acts as a fully automated battery management tool. As a result, their customers can lower their electricity costs and thereby support renewable energy integration through increased grid flexibility based on the flexibility of their battery capacity (Entrix Energy, 2024a).

Entrix's platform offers its serviced batteries access to several energy markets, trading both flexibility and grid stabilization services as well as energy trading on intra-day and day-ahead markets (Entrix Energy, 2024b). These markets have undergone significant changes throughout the last years, which explains how battery storage systems and, therefore, Entrix became a viable and possible business model. Most

significantly, structural changes in the flexibility markets have enabled Entrix's business model.

Batteries are mainly used in the FCR and automated Frequency Restoration Reserve (aFRR) market due to their short activation times. In 2021, they made up two-thirds of the German FCR capacity (Figgenger et al., 2022). The FCR market must immediately counteract grid deviations from 50 Hz $\pm$ 0.01 Hz, activated in up to 30 s. The capacity for offering FCR is currently sold in six 4-h blocks. If a battery producer's slot is acquired, in case of deviation, within these four hours, they must supply energy for an incident time frame of a minimum of 15 min. They have to stabilize the grid symmetrically, i.e., feed electricity into the grid or absorb it from it (Pant et al., 2024). This makes batteries, which can charge and discharge quickly, a perfect fit for FCR, which explains why they are used predominantly.

Changes to the FCR and aFRR market design have made participation in these significantly more attractive for battery services, giving birth to the market concepts leveraged by Entrix. The following sections first detail the key changes in FCR trading, followed by the changes in aFRR trading.

In FCR trading, first, the time slots in which FCR is traded were reduced from one month in 2011 to one day in 2019, down to a service period of four hours in July 2019. Further, the minimum capacity bid size was reduced from 5 MW to 1 MW. Finally, in 2019, the Bundesnetzagentur reduced the required incident timeframe over which electricity must be supplied from 30 to 15 min, thereby effectively halving the necessary battery capacity for participating in the FCR market and enabling small battery storage systems, which can be serviced by Entrix (Figgenger et al., 2022). Table 5 summarizes the market amendments, adapted from (Figgenger et al., 2022).

As a result, since 2017, the installed and qualified FCR capacity, provided by all technologies, including batteries, in Germany has increased by more than sevenfold, up to approximately 800 MW, as shown in Fig. 11. This marked increase in FCR capacity can also be explained by the increased share of renewable energies in the German electricity network, which commonly has weaker frequency inertia than their traditional, fossil-based generator counterparts (Schwarz and Cai, 2017). Thus, prices for FCR capacity exploded from below €1000/MW/week in 2021 to over €9000/MW/week in 2022 (Figgenger et al., 2022). In the long run, however, it can be expected that FCR capacity will be abundant as more battery capacity is installed for providing electricity (Poudineh and Apostolopoulou, 2024). FCR can be offered simultaneously, and hence, prices will decrease to a level where battery owners cannot finance their projects through FCR revenues alone (Pant et al., 2024). Regardless, we can conclude that the market liberalization in the FCR space gave rise to battery projects and business models based on their softwares optimization.

Second, when analyzing the changes in aFRR trading, one can conclude that market access for battery traders has become significantly easier. The aFRR is used to stabilize energy systems over a longer time-frame, with an activation time of 5 min (instead of 30 s for FCR) (ACER, 2022). Thus, aFRR follows FCR on a frequency incident recovery timeline. Here, too, power must be delivered for at least 15 min, but providers

Table 5
Changes in FCR market characteristics (Figgenger et al., 2022).

	Before July 2019	July 2019–July 2020	Since July 2020
Direction	Positive and negative power together	Positive and negative power together	Positive and negative power together
Minimal bid	1 MW	1 MW	1 MW
Minimal increment	1 MW	1 MW	1 MW
Reaction time	30 s	30 s	30 s
Provision time	15 min	15 min	15 min
Remuneration	Pay-as-bid for power	Market-clearing-price for power	Market-clearing-price for power
Time sectioning	1 week	24 h	4 h
Tendering	Tuesdays, 3 pm	D-2, 3 pm	D-1, 8 am
Demand Germany	551 MW–620 MW	620 MW	573 MW

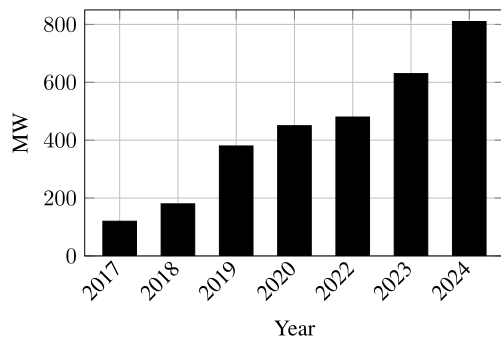


Fig. 11. Prequalified FCR BESS capacity in Germany (2017-2024) (Regelleistung.net, 2024).

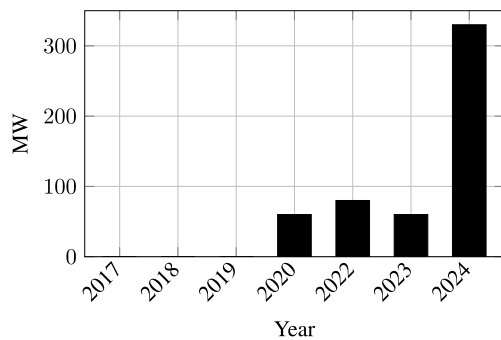


Fig. 12. Prequalified BESS capacity for aFRR in Germany (2017-2024) (Regelleistung.net, 2024).

must only either inject or remove energy from the grid. In that sense, the aFRR product is asymmetrical, whereas the FCR product is symmetrical. To increase the marketability of aFRR products, central European TSOs, including Germany developed and launched the PICASSO platform. It allows for the auctioning and activation of aFRR products that comply with a standard set of requirements. The platform went live in June 2022 with pioneering TSOs, coinciding with Entrix's founding date.

From 2020 to 2024 there is a clear jump in battery participation in the aFRR market observable. Remarkably, as illustrated in Fig. 12, from 2023 to 2024, installed aFRR battery capacity in Germany jumped from roughly 60 MW to 320 MW. While Entrix is likely not directly participating in PICASSO because the platform is focused on inter-TSO trading, it still facilitates the flexible trading of frequency reserve, which is part of Entrix's core business model. This aligns with the markedly increased battery volume in the aFRR market, which happened after 2022. Hence, the establishment of such a trading platform likely facilitated the functioning of Entrix's business model through improved inter-TSO marketability of aFRR.

In summary, the business model of Entrix has been greatly facilitated by increased liberalization of FCR market criteria, as well as increasingly automated and digitized market services in aFRR trading through the PICASSO.

6.2. Entelios

Entelios was founded in 2010 as Europe's first demand response provider. The company specializes in managing and marketing electricity flexibility to optimize energy costs and generate additional revenue for customers. The company's fully automated load management solutions integrate diverse assets such as industrial power-to-heat plants, battery storage, and flexible production processes, supporting decentralized energy systems and the energy transition (Entelios, 2024). Entelios provides flexibility across short-term markets, including balancing services (FCR, aFRR, mFRR) and the European power exchange EPEX SPOT,

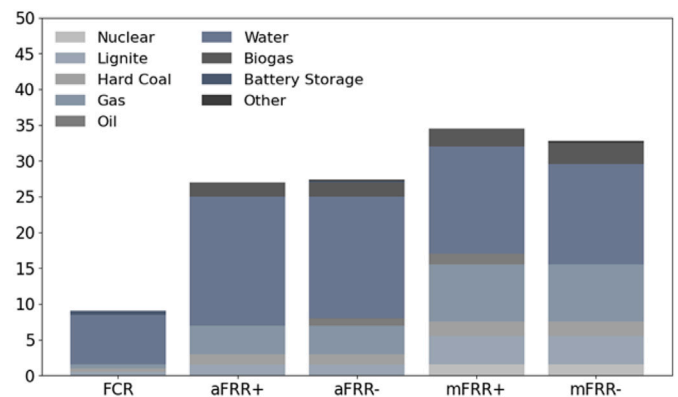


Fig. 13. Breakdown of prequalified capacity (GW) of German balancing service providers by energy carriers as of 2019 (Consentec GmbH, 2022).

delivering 1GW net flexibility - equivalent to one nuclear power plant (Business Wire, 2024; Entelios AG, 2023). It optimizes across markets, algorithmically maximizing available flexibility with real-time price signals and operational parameters. This enables customers to alleviate grid stress during peak demand while profiting from electricity markets (Entelios, 2024).

As a Balancing Service Provider (BSP), the company submits bids for balancing services in line with capacity requirements defined by Transmission System Operators (TSOs) (Belmonte et al., 2023). Historically, nuclear and coal plants played a central role in stabilizing the grid. Their phase-out, however, has increased the demand for alternative flexibility solutions, such as natural gas plants and large-scale batteries (Samseth et al., 2021). This trend is evident in the German case, illustrated in Fig. 13. This transition creates opportunities for renewable-focused BSPs (Belmonte et al., 2023) such as Entelios to leverage demand-side flexibility and renewables to support grid stability (Backer et al., 2023).

Furthermore, the German balancing market is shaped by both European and national guidelines. On a European level, the 2017 System Operation Guideline (SOG) harmonizes grid operations, ensuring stability and cross-border integration (Pedro et al., 2023). The Electricity Balancing Guideline (EBGL) further supports EU-wide market integration (Backer et al., 2023). Regulatory changes, such as the 2020 shift to four-hourly settlement prices for FCR and the 2019 removal of price caps for aFRR and mFRR, have increased flexibility and competition, further enabling players like Entelios to offer innovative demand-side solutions for grid stability (Belmonte et al., 2023).

On the other hand, Germany's balancing services and reserve energy framework is governed by the Electricity Grid Access Ordinance (StromNZV) and the EnWG, which establish rules for system responsibility and market-driven procurement. The StromNZV outlines reserve energy procurement and settlement, while the EEG enables renewable plants to participate. Providers must undergo a prequalification process mandated by the SOGL, which ensures competition and accounts for the differing technical requirements across market segments (Consentec GmbH, 2022).

Additionally, the AbLaV (Regulation on Interruptible Loads), introduced in 2012, enhances flexibility by allowing consumers to reduce or stop consumption during grid instability (Elspas et al., 2023). This regulatory framework significantly impacted Entelios, enabling the company to market flexible assets, including Germany's first remotely controlled paper machine (Entelios, 2024). It also demonstrates how the company aligns business growth with regulatory advancements.

The combination of these policies has lowered entry barriers and fostered competition, opening the door for players like Entelios to contribute to balancing services.

7. Case studies: downstream

Through an extensive literature review and research, it has become clear that the downstream segment offers the most accessible and diverse opportunities for entrepreneurs, particularly in customer-centric services and digital platforms. Innovation here can happen faster and scale more easily compared to upstream, which is more capital- and regulation-intensive. While midstream is growing in importance with the rise of smart grid solutions, downstream remains the prime space for new ventures to thrive. To illustrate this, the following section presents three case studies GridX, Enpal, and Naturstrom all operating within the downstream space but focusing on different aspects of the markets.

7.1. GridX

GridX was founded in May 2016 and specializes in integrating and managing DERs. In 2020, E.ON acquired a majority stake in GridX and later acquired the remaining shares, making it a fully owned subsidiary (GridX, 2024a). GridX offers a comprehensive approach, combining hardware, platform services, and expertise in energy system digital transformation. The company provides solutions for home energy management, offering users intuitive systems for managing household energy consumption (GridX, 2024e). GridX's product XENON connects, monitors, and controls various DERs, including batteries, EV chargers, and solar inverters (GridX, 2024b). The GridBox is a gateway to aggregate and process data in real-time, optimizing energy flows and ensuring system stability (GridX, 2024d). Additionally, XENON optimizes EV charging, improving cost-effectiveness and efficiency (GridX, 2024g).

Several market developments have supported GridX's growth journey. Growing DER adoption leads to a significant shift in power systems. These resources, often "behind-the-meter" and connected at low- and medium-voltage levels, introduce new dynamics into energy generation and demand (Birk et al., 2017; Guerrero et al., 2020). As DER adoption grows, it creates fluctuations in energy generation and demand, challenging the grid's stability. TSOs struggle with limited visibility into DER operations, which impacts the accuracy of load and generation forecasts. DSOs are taking on expanded roles to address these challenges. To prevent grid congestion and imbalances, they manage peak loads, ensure voltage stability, and support reactive power (GridX, 2024c). See Fig. 14 for an overview of responsibilities.

The efficient integration of DERs is essential for maintaining grid stability and minimizing costly infrastructure upgrades. DER management system unlocks demand-side resources, enhances flexibility, and makes renewable-powered grids more resilient and cost-effective. By incorporating grid signals into local energy management systems, DSOs can optimize DER performance and contribute to the overall energy balance (GridX, 2024c). XENON plays a vital role in this process. It offers real-time monitoring and control of decentralized resources, ensuring seamless integration into the grid.

Additionally, more and more energy consumers are becoming prosumers, which means they take on a more active role in energy production and usage. GridX's solutions offer real-time insights into energy flows, giving users greater transparency and control (GridX, 2024f). The EEG of 2017 facilitates this shift by supporting self-supply and energy communities (Inês et al., 2020). Prosumers benefit from tariffs designed to encourage efficient energy use, which GridX's XENON platform optimizes to lower costs and increase self-sufficiency.

Germany's energy infrastructure transformation was further accelerated in 2016 with the introduction of the Digitization of the Energy Transition Act (GDEW). The regulation focuses on establishing a cross-sector communication network, enhancing competition and interoperability in smart metering, and improving market communication via the Smart Meter Gateway (SMGW). The GDEW envisions metering point operators evolving into broader infrastructure service providers (EY, 2018). Its focus on smart grids and digital infrastructure makes technologies like GridX's key to optimizing decentralized energy resources in real-time.

Lastly, a key update of the EnWG, Section 14a (effective 1 January 2024), has recently influenced GridX. It allows grid operators to manage congestion by reducing power consumption from controllable systems like heat pumps, EV chargers, and residential batteries, offering reduced grid fees to households. This update helps prevent strain on the low-voltage grid and promotes active grid expansion and smart control solutions (GridX, 2024h). For GridX, this regulation highlights the growing need for advanced energy management systems that integrate and control diverse energy devices.

7.2. Enpal

Enpal's success is a direct result of energy policies like the EEG of 2000 and EnWG, which provided stability and incentives for renewable energy. The FiT system ensured predictable payments for solar energy, enabling Enpal to offer a no-upfront-cost solar subscription model. This lowered barriers for homeowners to adopt solar power, making renewable energy more accessible. The regulatory framework also allowed Enpal to innovate further by adding heat pumps and smart energy management solutions, showcasing how favorable policies can drive business growth and innovation in the energy sector.

Enpal, a Berlin-based company founded in 2017, provides homeowners with solar energy solutions through a subscription-based model. By offering solar panel installations without upfront costs, Enpal has played a key role in making renewable energy accessible to a broader audience (Arnold, 2024).

The energy market liberalization mandated equal grid access for renewable energy producers. Equal grid access helped to scale the deployment of solar power, increasing the attractiveness of solar power to energy consumers. As a result, companies like Enpal introduced innovative business models, such as solar leasing, to meet the financing demand for these solutions.

Several government policies provided the foundation for Enpal's emergence and rapid growth. The EEG of 2000 required grid operators to purchase electricity generated from renewable sources (Erge et al., 2001). It also introduced a FiT system that guaranteed fixed payments for solar energy fed into the grid for up to 20 years, ensuring financial viability for solar projects (Leiren and Reimer, 2020). These tariffs were financed via the EEG levy, which was imposed on nationwide electricity consumption. The FiT system significantly reduced the investment risks for private and commercial PV system operators. It provided the financial stability and trust needed for solar projects to flourish. Financial stability and confidence, more specifically, predictable cash flows, were critical to Enpal's leasing model.

Since its founding in 2017, Enpal has experienced impressive growth and is Germany's first green tech unicorn. By 2024, the company had installed solar systems for over 45,000 customers that collectively produce 345 GWh (PitchBook Data, Inc). The company extended its

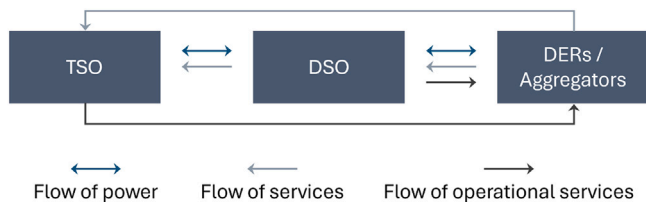


Fig. 14. Interaction model between system operators and DERs (Birk et al., 2017; GridX, 2024c).

offerings to include heat pumps and smart energy management solutions. With a valuation of \$2.6 billion, Enpal is one of Germany's largest and most impactful new energy ventures that emerged after the energy market liberalization (PitchBook Data, Inc).

The growth of Enpal is directly linked to policies such as the EnWG and the EEG. Both introduced measures to provide the stability necessary for PV investments. Most notably, the FiT system was an impactful measure to create trust among all stakeholders and to make PV systems economically viable.

7.3. Naturstrom

Naturstrom AG is a key case study for understanding the real effects of the EnWG and EEG on Germany's downstream energy sector. Its growth clearly shows how (i) the EnWG dismantled utility monopolies, enabling competition and opening the market to new entrants, and (ii) the EEG ensured a stable and viable supply of green electricity, which was essential for sustainability-focused retailers like Naturstrom to succeed. Together, these factors demonstrate how policy reforms directly led to greater consumer choice and new business opportunities in the energy market.

Founded in 1998, Naturstrom AG is one of Germany's first independent energy retailers specializing in 100 % renewable electricity. Today, the company is active in energy generation and supply. It supplies green energy from wind, solar, and hydroelectric power to end customers. Unlike traditional utilities, Naturstrom has positioned itself as a sustainability-focused alternative for environmentally conscious consumers. Its ability to operate as an independent retailer marked a significant departure from the monopolistic structure of the pre-liberalization energy market (Naturstrom AG, 2024).

The EnWG made the emergence of Naturstrom possible. The dismantling of regional monopolies, which had previously restricted energy suppliers to certain geographical locations, and non-discriminatory third-party grid access meant that Naturstrom was now able to sell electricity to customers throughout Germany, regardless of their location. Consumers were given the ability to choose their energy supplier freely. Fig. 15 illustrates this development. Between 2007 and 2023, the share of network areas with more than 100 active suppliers increased from 3.7 % to 73.3 %, which indicates an increase in market diversity and competition. In contrast, areas served by fewer than 50 suppliers shrank significantly, indicating the erosion of monopolistic structures. As a result, Naturstrom was able to scale its operations and reach a national customer base, which had been impossible in the pre-liberalized market dominated by regional utilities.

Naturstrom's growth and diversification were strongly shaped by two key legislative frameworks in Germany. (i) The EEG (2000) played a pivotal role in establishing a viable market for green electricity by mandating priority grid access for renewables and introducing feed-in tariffs, which guaranteed fixed payments to producers and significantly

increased the availability of clean energy. This expanded supply enabled Naturstrom to secure and deliver greater volumes of green electricity, strengthening its position as a clean energy retailer. (ii) The EnWG further accelerated Naturstrom's expansion by removing geographic restrictions and ensuring non-discriminatory, nationwide grid access for independent retailers. This allowed Naturstrom to compete directly with incumbent utilities and reach a broader customer base. As a result of these supportive policies, Naturstrom's green electricity customers grew from over 100,000 in 2010 to more than 300,000 by 2024, while the company also diversified into biogas, decentralized energy consulting, and local energy production projects (Naturstrom AG, 2024).

8. Conclusion

This study examines the impact of energy market liberalization and renewable energy policies on the emergence of new energy ventures within the electricity sector over the past 25 years in Germany. Using a systematic research framework, the study categorizes these ventures based on both technology type and value chain positioning. The key findings are as follows:

- *Policy-driven growth:* The analysis suggests a positive association between policy measures (such as the EnWG, EEG, and GEG) and the emergence of new energy ventures. While various factors contribute to this growth, these policies appear to have played an important enabling role by opening markets, encouraging competition, and supporting the expansion of renewable energy supply.
- *Downstream dominance:* A closer review of the ventures shows that the downstream segment offers the most accessible and diverse opportunities for entrepreneurs. This is especially true in customer-centric services and digital platforms, where innovation tends to happen faster and scale more easily compared to the capital and regulation-intensive upstream segment. While midstream is growing in relevance, particularly with the need for grid upgradation/reinforcement and ancillary services.
- *Emerging opportunities:* BESS, heat pumps, smart metering, and energy management solutions like HEMS and grid management software represent the most prominent emerging business areas at present. The upcoming grid inertia market in Germany, planned for launch in January 2026, is expected to attract further entrepreneurial activity in the BESS and flexibility services domains.
- *Technology is not enough:* The study finds that technological availability alone is insufficient for market adoption. For example, the widespread uptake of heat pumps and HEMS in Germany has relied heavily on policy incentives and subsidies that influence consumer behavior.

While this study provides a detailed account of venture creation during Germany's energy transition, its conclusions are shaped by the following limitations.

8.1. Limitations of the study

First, our scope is limited to firm emergence rather than performance. We track the creation of new ventures in response to policy but do not assess their long-term viability or economic impact. This distinction is crucial: we demonstrate a link to entrepreneurial activity, but not necessarily to sustainable commercial success.

Second, our causal claims are based on a qualitative analysis of temporal patterns. We connect the emergence of new business models to a detailed timeline of policy events, highlighting compelling associations between them. While this approach provides rich contextual insights, it is not designed to statistically isolate policy impacts from other concurrent economic or technological developments. Therefore, our causal assertions should be understood as evidence-based hypotheses, not as statistically proven relationships.

Finally, while the study's conclusions are drawn from Germany's specific institutional setting, they offer a broadly relevant analytical

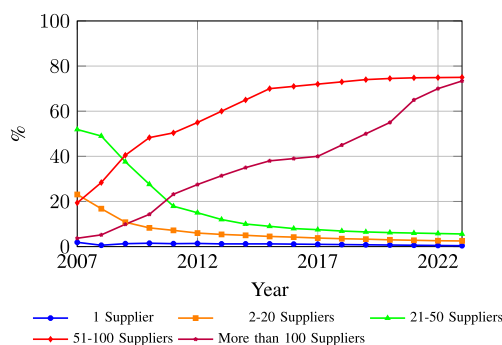


Fig. 15. Proportion of network areas in which the number of suppliers shown is active (2007–2023) (Bundesnetzagentur, 2010–2023).

framework. The German Energiewende serves as a powerful case study, but rather than providing a directly replicable blueprint, this research identifies key principles and dynamics. These insights are intended to be informative for other nations, helping policymakers understand the conditions needed to foster similar entrepreneurial ecosystems.

8.2. Future research directions

Building on the findings of this study, future research should focus on three areas:

- *Scalability in policy-shaped markets:* Further investigation is needed to understand how policy frameworks and market dynamics affect the scalability of new ventures, particularly in downstream energy segments, which have low entry barriers and require less capital. Research should identify which regulatory mechanisms foster sustainable growth versus those that might lead to rapid market entry followed by premature saturation, increasing the risk of venture or even market failures.
- *Venture failure analysis in regulated energy markets:* Beyond success stories, systematic research is crucial to understand why some energy ventures fail in environments with significant regulatory oversight. This includes identifying patterns in business model deficiencies, misalignments with regulatory frameworks, and errors in market timing. Such insights are vital for developing more resilient business strategies.
- *Impact of divergent drivers on energy ventures:* Future research should examine how energy ventures evolve under proactive policy

incentives, often seen in the Global North, versus reactive responses to systemic challenges, common in the Global South. The proposed analysis should assess impacts on innovation pathways, business model viability, and the transferability of strategies across diverse socio-economic and infrastructural contexts.

CRediT authorship contribution statement

Prashant Pant: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Frank-Martin Belz:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the TUM SEED Center, as part of the exceed program funded by the German Academic Exchange Service (DAAD) and the German Federal Ministry for Economic Cooperation and Development (BMZ).

Appendix

See Table A1 and Fig. A1.

Table A1

Number of energy companies by segment and year.

Market Segment	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Transmission System Operators										4	4	4	4	4
Transmission Network Operators										16	15	16	15	14
Electricity Distribution Network Operators	910	910	920	940	940	930	920	906	907	908	903	898	889	887
Electricity Generators										0	1126	1143	1156	1177
Electricity Generators (> 100 MW)	300	300	300	300	300	70	70	90	80	87	96	0	0	0
Electricity Storage Operators (> 1 MW or > 1MWh)										107	123	137	110	109
Electricity Traders (Balance Responsible Parties)	120	120	140	130	140	130	130	130	55	57	60	1277	1102	1122
Electricity Suppliers	1030	1050	1100	1170	1190	1190	1190	1200	1280	1327	1350	1364	1359	1350
Natural Gas Production Companies	9	4	5	6	6	7	7	7	8	8	7	7	7	7
Gas Distribution Network Operators	710	720	720	730	730	730	720	730	733	716	722	711	704	706
Gas Storage Operators	10	18	23	24	27	29	40	41	25	31	30	31	31	31
Gas Wholesalers (Balance Responsible Parties)										0	562	564	355	366
Gas Traders	40	60	60	70	70	70	70	70	67	68	65	0	0	0
Gas Wholesalers	800	820	850	890	890	910	910	930	970	1014	1049	1051	1038	1026
District Heating/Cooling Producers	470	520	530	540	550	550	550	540	560	597	598	602	609	626
District Heating/Cooling Network Operators	340	410	410	420	430	430	430	450	450	454	457	461	465	475
District Heating/Cooling Suppliers	460	550	560	580	580	580	600	540	550	561	564	566	573	590
Heat Storage Operators										20	21	44	45	54
	1700	1800	1900	2000	2000	2000	2000	2000	2000	2225	2234	2300	2297	2295

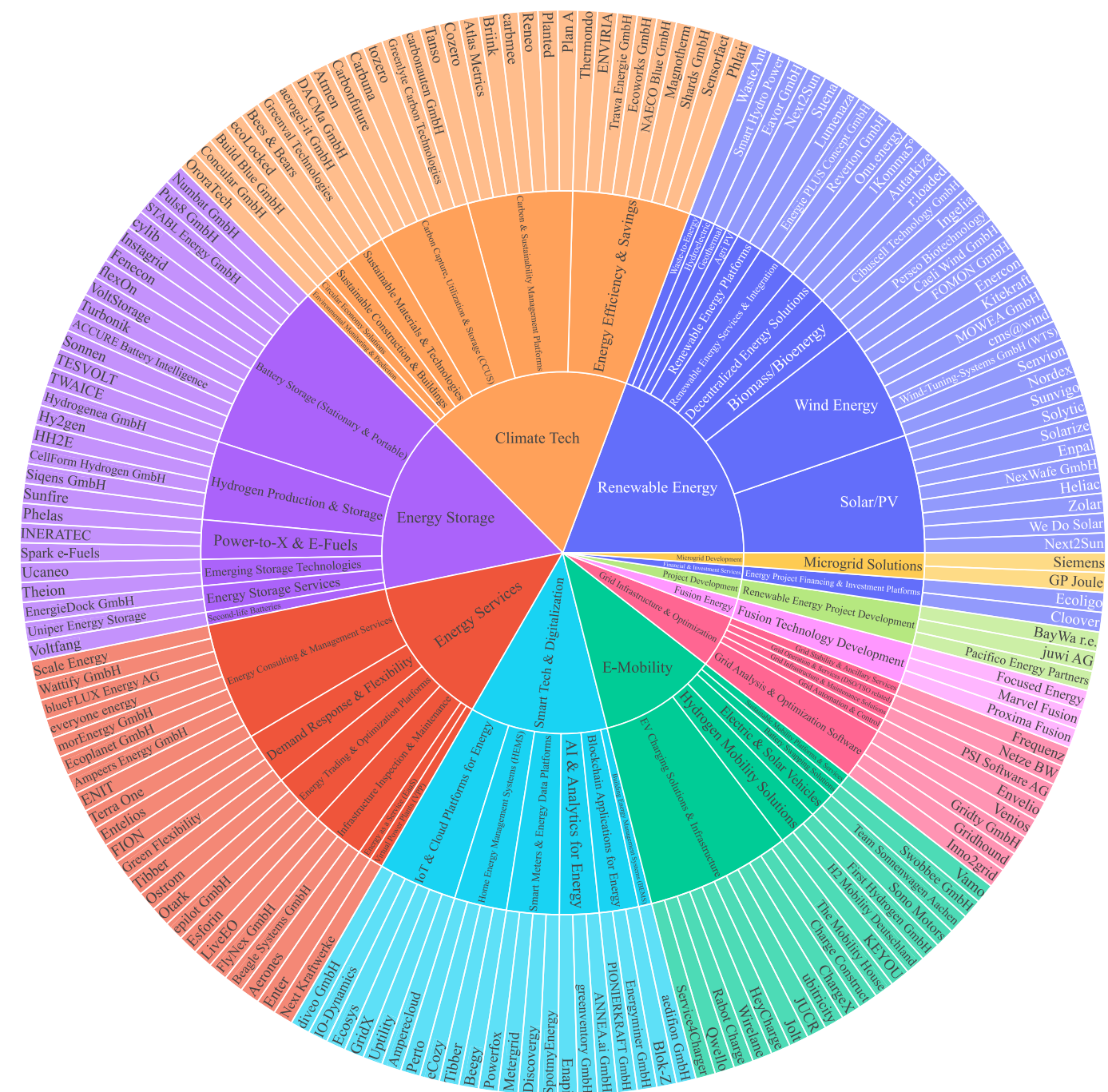


Fig. A1. Simplified visualization of the German energy ecosystem (240+ companies); complete data and code are available in the Git repository (Prashantgrid, n.d.).

Data availability

No data were used for the research described in the article.

References

- ACER, 2022, Sep. Decision No. 15/2022 on the amendment of the aFRR implementation framework. ENTSO-E, Tech. Rep. [Online]. Available: https://eepublicdownloads.entsoe.eu/clean-documents/nc-tasks/220921_ACER%20Decision%2015-2022%20on%20the%20aFRRIF%20-%20Annex%20II.pdf (Accessed: 7-Dec-2024).
- Agora Energiewende, 2016. Energiewende: what do the new Laws mean? Ten questions and answers about EEG 2017, the electricity market Act, and the digitisation Act. <https://www.agora-energiewende.de/en/publications/energiewende-what-do-the-new-laws-mean/>
- Ahlemeyer, K., Griesse, K.-M., Wawer, T., Siebenhüner, B., 2022. Success factors of citizen energy cooperatives in north western Germany: a conceptual and empirical review. *Energy Sustain. Soc.* 12, 29.
- AI Research Tool, 2022, Aug. Summary of: IMPROVED advises GreenCom networks - a leading Germany based energy management solution provider - on the sale to Enphase energy (alternative summary), Web page summary, [Online]. Available: <https://improvedcf.com/wp-content/uploads/2022/09/220829-IMPROVED-Greencom-Press-Release-Final.pdf>
- Al-Ali, A.-R., Zulkarnan, I.A., Rashid, M., Gupta, R., AliKarrar, M., 2017. A smart home energy management system using IoT and big data analytics approach. *IEEE Trans. Consum. Electron.* 63 (4), 426–434.
- Amprion GmbH, 2022. Amprion innovation report. Amprion GmbH, Tech. Rep., [Online]. Available: <https://www.amprion.net/documents/Amprion/Innovation/Amprion-Innovation-Report.pdf> (Accessed: 2025-May-12).
- Anchustegui, I.H., 2018. Transmission Networks in Electricity Competition: Third-Party Access and Unbundling – A Transatlantic Perspective, Competencia en mercados con

- recursos esenciales compartidos: telecomunicaciones y energía. University of Valencia, forthcoming (2018). [Online]. Available: <https://ssrn.com/abstract=3159458>
- Aniello, G., Többen, J., Kuckshinrichs, W., 2019. The transition to renewable energy technologies—impact on economic performance of North Rhine-Westphalia. *Appl. Sci.* 9 (18), 3783.
- Arnold, T., 2024. Wie das grüne Einhorn Enpal mit Solarenergie Gewinne schreibt. <https://www.handelsblatt.com/unternehmen/energie/start-up-wie-das-gruene-einhorn-enpal-mit-solarenergie-gewinne-schreibt/29265468.html>
- Backer, M., Keles, D., Kraft, E., 2023. The economic impacts of integrating European balancing markets: the case of the newly installed aFRR energy market-coupling platform PICASSO. *Energy Econ.* 128, 107124.
- BDEW - Bundesverband der Energie- und Wasserwirtschaft e.V., 1998–2024. Non-Public Dataset on Energy Companies, Provided by BDEW via email communication. Data received: 15-Dec-2024.
- Belmonte, B.B., Mouratidis, P., Franke, G., Rinderknecht, S., 2023. Developments in the cost of grid balancing services and the design of the European balancing market. *Energy Rep.* 10, 910–931.
- Bendig, D., Brüß, L., Degen, F., 2025. Entrepreneurship in the renewable energy sector: a systematic literature review of types, characteristics, and sustainability impacts. *Renew. Sustain. Energy Rev.* 212, 115337. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1364032125000103>
- Berger, S., Ebeling, F., Feldhaus, C., Löschel, A., Wyss, A.M., 2022. What motivates smart meter adoption? Evidence from an experimental advertising campaign in Germany. *Energy Res. Soc. Sci.* 85, 102357. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2214629621004485>
- Birk, S., Fleer, J., Holtz, G., Jeddi, S., Lilienkamp, A., Namockel, N., Schönfisch, M., Wagner, J., 2021 Apr. Geschäftsmodelle für Regionale Virtuelle Kraftwerke. Energiewirtschaftliches Institut an der Universität zu Köln gGmbH (EWI) and Technische Hochschule Köln - Cologne Institute for Renewable Energy (CIRE) and Forschungszentrum Jülich - Institut für Energie- und Klimaforschung - Systemforschung und Technologische Entwicklung (IEK-STE) and Wuppertal Institut für Klima, Umwelt, Energie - Zukünftige Energie- und Industriesysteme, Tech. Rep., [Online]. Available: https://www.ewi.uni-koeln.de/cms/wp-content/uploads/2022/04/210420_VISE_Endbericht_v1.0.pdf
- Birk, M., Chaves-Ávila, J.P., Gómez, T., Tabors, R., 2017. TSO/DSO coordination in a context of distributed energy resource penetration. *Proc. EEIC, MIT Energy Initiative Rep.*, Cambridge, MA, USA, pp. 2–3.
- Bundesministerium für Wirtschaft und Klimaschutz (BMWK). Bundesförderung für effiziente Gebäude (BEG). <https://www.bmwk.de/Redaktion/DE/Dossier/bundesfoerderung-fuer-effiziente-gebäude.html>
- Brunkreeft, G., 2000. Regulatory issues in liberalised electricity markets: theory and case studies. *Utilities Policy* 9 (3), 163–178.
- Bundesnetzagentur, Bundeskartellamt, 2010–2023. Monitoringbericht series (2010–2023). Data referenced from editions 2010, 2015, 2020, and 2023. [Online]. Available: <https://www.bundesnetzagentur.de> (Accessed: 10 Dec 2024)
- Bundesnetzagentur, 2015 Nov. Festlegung zur Spezifizierung der Regelungen für die Beschaffung von Regelleistung und zur Veröffentlichung von Informationen über die Regelungen von TenneT TSO GmbH, Amprion GmbH, 50Hertz Transmission GmbH und TransnetBW GmbH. Bundesnetzagentur, Bonn, Beschluss BK6-15-097.
- Bundesnetzagentur, 2024. History of liberalisation. [Online]. Available: <https://www.bundesnetzagentur.de/EN/Areas/Energy/GeneralInformationRegulation/HistoryOfLiberalisation/start.html> (Accessed: 2-Dec-2024)
- Bundesnetzagentur, 2024. Integration von steuerbaren Verbrauchseinrichtungen. https://www.bundesnetzagentur.de/DE/Home/home_node.html (Accessed on 18 Nov 2024.)
- Bundesregierung, 2023. Ausbau erneuerbarer Energien massiv beschleunigen. [Online]. Available: <https://www.bundesregierung.de/breg-de/service/archiv/novelle-eeg-gesetz-2023-2023972>
- Burger, S., Jenkins, J.D., Battie, C., Pérez-Arriaga, I.J., 2019. Restructuring revisited—Part 1: a proposed new market paradigm. *Energy J.* 40 (3), 1–26.
- Burger, B., 2024. Öffentliche Nettostromerzeugung in Deutschland im Jahr 2023. <https://www.energy-charts.info/?l=en&c=DE>
- Business Wire, 2024. Brennt Energy kooperiert mit Entelios, um nachhaltige Energiepraktiken einzuführen und so die Einnahmequellen für bGen-Anwender in Deutschland zu optimieren. [Online]. Available: <https://shorturl.at/taEmB> (Accessed: 6-Dec-2024).
- Böhlinger, C., Landis, F., Reaños, M.A.T., 2017. Economic impacts of renewable energy promotion in Germany. *Energy J.* 38 (1 suppl), 189–210. [Online]. Available: <https://doi.org/10.5547/01956574.38.S11.cb0h>
- Bürer, M.J., Wüstenhagen, R., 2009. Which renewable energy policy is a venture capitalist's best friend? Empirical evidence from a survey of international cleantech investors. *Energy Policy* 37 (12), 4997–5006.
- Calliao, K., 2024. IKOMMA5° announces record sales and profitable growth. <https://1komma5.com/au/press/press-releases/1komma5-announces-record-sales-and-profitable-growth/> Updated: 27 February 2024. (Accessed: 12 May 2025).
- Chatterjee, P., Petitjean, O., Perez, A., Steinfort, L., 2023 Nov. Green' multinationals exposed: how the energy transition is being hijacked by corporate interests. Transnational Institute (TNI), Tech. Rep., Available online: <https://tni.org/GreenMultinationals>
- Clearingstelle EEG/KWK, 2023. Arbeitsausgabe der Clearingstelle EEG—KWK. Gesetz für den Ausbau erneuerbarer Energien. [Online]. Available: https://www.clearingstelle-eeg-kwk.de/sites/default/files/2023-02/EEG-230201-230104-web_0.pdf (Accessed: 22-Dec-2024).
- Consentec GmbH, 2022 Aug. Beschreibung von Konzepten des Systemausgleichs und der Regelreservemärkte in Deutschland: Erläuterungsdokument im Auftrag der deutschen Übertragungsnetzbetreiber. Consentec GmbH, Aachen, Germany, Report.
- Deutsche Energie-Agentur (dena), 2024. Germany's regulatory framework and instruments for energy efficiency in industry. https://www.dena.de/fileadmin/dena/Dokumente/Projektportrait/Projektarchiv/Entrans/Germany_s_Regulatory_Framework_and_Instruments_for_Energy_Efficiency_in_Industry_EN.pdf (Accessed: 2025-May-13).
- Deutscher Bundestag, 2010 Jul. Gesetz zur Änderung des Rechtsrahmens für Strom aus solarer Strahlungsenergie und zu weiteren Änderungen im Recht der erneuerbaren Energien. *Bundesgesetzblatt (BGBl.) Teil I*, pp. 983–987.
- Deutscher Bundestag, 2022. Erneuerbare-Energien-Gesetz (EEG 2023), BGBl. I, S. 1722, 2022. Available at: https://www.gesetze-im-internet.de/eeg/2014/EEG_2023.pdf
- E. S. News, 2023. Sonnen's residential virtual power plant hits 250MWh capacity in Germany. [Online]. Available: <https://www.energy-storage.news/sonnen-residential-vpp-Germany/> (Accessed: 2025-May-14).
- E. S. News, 2025. Partners group invests €400 million in German battery developer green flexibility. [Online]. Available: <https://www.energy-storage.news/partners-group-green-flexibility/> (Accessed: 2025-May-14).
- E.ON SE, 2024. E.ON at a glance. [Online]. Available: <https://www.eon.com/en/about-us.html> (Accessed: 2-Dec-2024).
- E.ON SE, 2018 Jun. E.ON completes sale of uniper stake. [Online]. Available: <https://www.eon.com/en/about-us/media/press-release/2018/2018-06-26-eon-completes-sale-of-uniper-stake.html> (Accessed: 2-Dec-2024).
- E.ON SE, 2024. E.ON history. [Online]. Available: <https://www.eon.com/en/about-us/e-on-history.html>
- E.ON. On course for net zero, Third Edition, March 2023, p. 20. Available at: <https://www.eon.com/en/press/publications.html>
- Eavor Technologies Inc. Year in review 2023, Webpage, [Online]. Available: <https://www.eavor.com/year-in-review-2023/> Accessed: 2025-May-12).
- ecostor, 2025. Eco stor AS announces contract for delivery of 10MW / 12MWh battery energy storage system. [Online]. Available: <https://www.eco-stor.com/resources/news/eco-stor-as-announces-contract-for-delivery-of-10mw-12mwh-battery-energy-storage-system>
- Elspas, M., GraSection Mann, N., Rasbach, W. ed), 2023. EnWG Energiewirtschaftsgesetz mit AbLaV, ARegV, GasGVV, GasHDRLtV, GasNEV, GasNZV, KAV, KraftNAV, LSV, MaStRV, NAV, NDAV, NetzResV, StromGVV, StromNEV, StromNZV, SysStabV, ÜNSchutzV, WasserstoffNEV – Kommentar, 2nd ed. Erich Schmidt Verlag, Berlin.
- Enerparc AG, 2024. Enerparc – pioneer in solar power. [Online]. Available: <https://www.enerparc.de/en> (Accessed: 3-Dec-2024).
- Englberger, S., Jossen, A., Hesse, H., 2020. Unlocking the potential of battery storage with the dynamic stacking of multiple applications. *Cell Rep. Phys. Sci.* 1 (11), 100238. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2666386420302563>
- Enpal GmbH, 2024. Solaranlage mieten: Lohnt sich das Modell? Describes Enpal's solar rental model, including no upfront costs and all-inclusive service. [Online]. Available: <https://www.enpal.de/photovoltaik/solaranlage-mieten>
- Entelios AG, 2023. Wer ist Entelios? YouTube. [Online]. Available: <https://www.youtube.com/watch?v=55bAR-j6QHE> Accessed: 6-Dec-2024).
- Entelios, 2024. Entelios – Lösungen für Energie und Flexibilität. [Online]. Available: <https://www.entelios.de/> (Accessed: 7-Dec-2024).
- Entelios, 2024. Über uns. [Online]. Available: <https://www.entelios.de/ueber-uns/> (Accessed: 7-Dec-2024).
- Entrix Energy, 2024a. Solution. <https://www.entrixenergy.com/de/solution/> (Accessed Aug 29, 2025)
- Entrix Energy, 2024b. Home. <https://www.EntrixEnergy.com/de/home/> (Accessed Aug 29, 2025)
- ENTSO-E, 2017 Jun. Consultation report “FCR cooperation” - results of the public consultation on potential market design evolutions for the FCR cooperation. ENTSO-E (European Network of Transmission System Operators for Electricity). Tech. Rep.
- Erge, T., Hoffmann, V.U., Kiefer, K., 2001. The German experience with grid-connected PV-systems. *Solar Energy* 70 (6), 479–487.
- European Commission, 2017 Nov. Commission regulation (EU) 2017/2195 of 16 November 2017 establishing a guideline on electricity balancing. Off. J. Eur. Union, 6–53. Commonly known as the Electricity Balancing Guideline (EBGL). Entered into force December 18, 2017. [Online]. Available: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32017R2195>
- European Committee for Electrotechnical Standardization (CENELEC), 2019. EN 50549-1:2019 – Requirements for Generating Plants to Be Connected in Parallel with Distribution Networks – Part 1: Connection to a LV Distribution Network – Generating Plants Up to and Including Type B. European Standard, Available at: <https://www.cenelec.eu/>
- EY, 2018. Barometer Digitalisierung der Energiewende: Ein neues Denken und Handeln für die Digitalisierung der Energiewende. [Online]. Available: <https://www.bmwk.de/Redaktion/DE/Publikationen/Studien/barometer-digitalisierung-der-energiewende.pdf> (Accessed: 10-Dec-2024).
- Facchinetti, E., 2022. Eleven business opportunities emerging from the energy transition. *Fatras, N., Ma, Z., Duan, H., Jørgensen, B.N., 2022. A systematic review of electricity market liberalisation and its alignment with industrial consumer participation: a comparison between the Nordics and China. Renew. Sustain. Energy Rev.* 167, 112793.
- Figgener, J., Tepe, B., Rücker, F., Schoeneberger, I., Hecht, C., Jossen, A., Sauer, D.U., 2022. The influence of frequency containment reserve flexibilization on the economics of electric vehicle fleet operation. *J. Energy Storage* 53, 105138.
- FOCUS Business, 2020. Next Kraftwerke is one of the FOCUS growth champions 2020. FOCUS Business, <https://www.focus.de/finanzen/boerse/boerse-exklusiv/ranking-fokus-wirtschaft-die-aufsteiger-des-jahres-2020>
- Geels, F.W., 2002. Technological transitions as evolutionary reconfiguration processes: a multi-level perspective and a case-study. *Res. Policy* 31 (8–9), 1257–1274.

- Gesetz zur Einsparung von Energie und zur Nutzung erneuerbarer Energien zur Wärme- und Kälteerzeugung in Gebäuden (Gebäudeenergiegesetz - GEG), PDF-Version des Gebäudeenergiegesetzes (Stand: aktuell). [Online]. Available: <https://www.gesetze-im-internet.de/geg/GEG.pdf>
- German Federal Government, 2010 Aug. Erstes Gesetz zur Änderung des Erneuerbare-Energien-Gesetzes (sog. PV-Novelle), Bundesgesetzblatt (BGBl.) I, p. 1078, Sixth Amendment to the EEG 2009, effective August 11, 2010. As referenced by Clearingstelle EEG|KWKG. <https://www.clearingstelle-ee-g-kwkg.de/gesetz/967>. (Accessed 14 May 2024)
- Germany Trade & Invest Gesellschaft für Außenwirtschaft und Standortmarketing mbH, 2023 Feb. Unlocking Germany's hidden highlights, Germany Trade & Invest Gesellschaft für Außenwirtschaft und Standortmarketing mbH, Tech. Rep. [Online]. Available: <https://www.gtai.de/resource/blob/61342/0517d51d8106982c4b0e858a506d3a82/studie-unlocking-Germany-s-hidden-highlights-data.pdf> (Accessed on 13 May 2025).
- Golly, O., Könnecke, L., Unger, B., 2025. German heating market growth boosts heat contractors, Arthur D. Little, viewpoint. [Online]. Available: https://www.adlittle.com/sites/default/files/viewpoints/ADL_German_heating_market_growth_2025.pdf. (Accessed on 13 May 2025).
- GridX, 2024a. About us. [Online]. Available: <https://www.gridx.ai/de/about> (Accessed: 10-Dec-2024).
- GridX, 2024b. Xenon – Die Plattform für Energiemanagement. [Online]. Available: <https://www.gridx.ai/de/xenon> (Accessed: 10-Dec-2024).
- GridX, 2024c. What is a grid operator? [Online]. Available: <https://www.gridx.ai/knowledge/what-is-a-grid-operator> (Accessed: 11-Dec-2024).
- GridX, 2024d. Edge Services – Lokale Optimierung für dezentrale Energiesysteme. [Online]. Available: <https://www.gridx.ai/de/edge-services> (Accessed: 10-Dec-2024).
- GridX, 2024e. Home Energy Management System – Intelligente Energie für Zuhause. [Online]. Available: <https://www.gridx.ai/de/use-cases/home-energy-management-system> (Accessed: 11-Dec-2024).
- GridX, 2024f. Was ist ein Prosumer? [Online]. Available: <https://www.gridx.ai/de/knowledge/prosumer> (Accessed: 10-Dec-2024).
- GridX, 2024g. Smart Charging – Intelligentes Laden für Elektrofahrzeuge. [Online]. Available: <https://www.gridx.ai/de/use-cases/smart-charging> (Accessed: 11-Dec-2024).
- GridX, 2024h. Paragraph 14a EnWG – Was steckt dahinter? [Online]. Available: <https://www.gridx.ai/knowledge/paragraph-14a-enwg> (Accessed: 10-Dec-2024).
- GridX, 2024j. HEMS report 2024: Home energy management systems in europe. Available at: <https://www.gridx.ai/resources/hems-report-2024>
- Guerrero, J., Gebbran, D., Mhanna, S., Chapman, A.C., Verbič, G., 2020. Towards a transactive energy system for integration of distributed energy resources: home energy management, distributed optimal power flow, and peer-to-peer energy trading. *Renew. Sustain. Energy Rev.* 132, 110000.
- Haelg, L., 2017. Policy Mixes and Their Influence on Innovation: A Longitudinal Analysis of the German and Swiss Solar Photovoltaic Sectors (Ph.D. dissertation). ETH Zurich. [Online]. Available: <https://doi.org/10.3929/ethz-b-000210222>
- Hirth, L., Glotzbach, S., 2021. The new redispatch regime in Germany (redispatch 2.0): integrating distributed generation into congestion management. *Energy Policy* 158, 112546.
- International Energy Agency, 2020. Status of power system transformation 2020. Tech. Rep. OECD/IEA, Paris.
- International Energy Agency, 2022 Jun. Unlocking the potential of distributed energy resources. International Energy Agency (IEA), Paris, France, Report, [Online]. Available: <https://www.iea.org/reports/unlocking-the-potential-of-distributed-energy-resources>
- International Energy Agency, 2024. Renewables 2024. Tech. Rep. OECD/IEA, Paris.
- Inés, C., Guilherme, P.L., Esther, M.-G., Swantje, G., Stephen, H., Lars, H., 2020. Regulatory challenges and opportunities for collective renewable energy prosumers in the EU. *Energy Policy* 138, 111212.
- IWR Pressedienst, 2021 Feb. Venios und Solandeo schaffen professionelle Lösung für Redispatch 2.0 und Ausbaustufen. <https://www.iwrpressedienst.de/energie-themen/pm-6959-venios-und-solandeo-schaffen-professionelle-loesung-fuer-redispatch-2-0-und-ausbaustufen>
- Jacobson, M.Z., 2021. The cost of grid stability with 100 clean, renewable energy for all purposes when countries are isolated versus interconnected. *Renew. Energy* 179, 1065–1075.
- Joskow, P.L., 2008. Lessons learned from electricity market liberalization. *Energy J.* 29 (SI2), 9–42.
- Kappner, K., Letmathe, P., Weidinger, P., 2023. Causes and effects of the German energy transition in the context of environmental, societal, political, technological, and economic developments. *Energy Sustain. Soc.* 13 (1), 28. [Online]. Available: <https://doi.org/10.1186/s13705-023-00407-2>
- KfW Bankengruppe, 2024. KfW program 270 – renewable energies: standard. Available at: <https://www.kfw.de/> (Accessed 15 May 2025).
- Klein, C., Düllmann, P., Osterkamp, L., Corte, N., Ruffing, P., Huq, I., Ahmedi, A., Yates, D., Leterme, W., 2023. Black start of HVDC links via grid-forming Wind Turbines: sensitivity analysis and technical requirements. In: *Renewable Energy Grid Integration Week 2023*. [Online]. Available: <https://conference-service.com/regridweek2023/documents/track2.html>. Accessed: 2025-May-12).
- Kotilainen, K., Saari, U.A., 2018. Policy influence on consumers' evolution into prosumers—empirical findings from an exploratory survey in Europe. *Sustainability* 10 (1), 186.
- Leiren, M.D., Reimer, I., 2020. Germany: from feed-in tariffs to greater competition. In: *Comparative Renewables Policy*. Routledge, pp. 75–102.
- Lenz, K., Lenck, T., Peter, F., 2019 Dec. The liberalisation of electricity markets in Germany: history, development and current status. *Agora Energiewende*, Berlin, Germany, Tech. Rep., version 1.0. Available online: <https://www.agora-energiewende.de>
- Liebensteiner, M., Ocker, F., Abuzayed, A., 2025. High electricity price despite expansion in renewables: how market trends shape germany's power market in the coming years. *Energy Policy* 198, 114448. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0301421524004683>
- Lietz, F., 2023 Nov. Bald Erleichterungen für die Geothermie? Webpage. [Online]. Available: <https://rgc-news.de/post/2617/bald-erleichterungen-fuer-die-geothermie> (Accessed: 2025-May-12).
- LoSection Ner, M., Böttger, D., Bruckner, T., 2017. [Online]. Available: Economic assessment of virtual power plants in the German energy market — a scenario-based and model-supported analysis. *Energy Econ.* 62, 125–138. <https://www.sciencedirect.com/science/article/pii/S0140988316303541>
- Lux, T., 2024. Enerparc: how Stefan Müller is shaping the solar market, Podcast, [Online]. Available: <https://omr.com/de/daily/enerparc-solar-stefan-mueller-omr-podcast> (Accessed: 3-Dec-2024).
- Mao, T., Guo, X., Xie, P., Zhou, J., Zhou, B., Han, S., Wu, W., Sun, L., 2020. Virtual power plant platforms and their applications in practice: a brief review. In: *2020 IEEE Sustainable Power and Energy Conference (ISPEC)*, pp. 2071–2076.
- Mario, R., 2005. Zusammenfassende Analyse zu Effektivität und ökonomischer Effizienz von Instrumenten zum Ausbau der Erneuerbaren Energien im Strombereich. Fraunhofer-Institut für System- und Innovationsforschung ISI, Karlsruhe, Germany, Endbericht.
- Marş, Z., Silva, C., Vale, Z., 2020. The key role of aggregators in the smart grid ecosystem: a review of current literature. *Int. J. Electr. Power Energy Syst.* 117, 105610.
- McKinsey & Company, 2022 Nov.. Fueling the energy transition: opportunities for financial institutions. <https://www.mckinsey.com/industries/financial-services/our-insights/fueling-the-energy-transition-opportunities-for-financial-institutions> (Accessed: 2025-May-6)
- Menrad, K., Nov. 2015. Virtual power Plants: contributions to flexibility? In: *DFBEE Conference: Flexibilisation of the Power System in France and Germany*, Franco-German Office for Renewable Energies (DFBEE), Berlin, Germany. Presentation at the DFBEE Conference.
- Mihailova, D., 2023. Redefining business models for the energy transition: social innovation and sustainable value creation in the European energy system. *Energy Res. Soc. Sci.* 100, 103114. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2214629623001743>
- Mouritz, M., Silva, A., Economou, D., Hargroves, C., Inchauspe, J., Akimov, A., Stegen, S., Nutkani, I., Vahidnia, A., McGrath, B., 2022. Business Models and the Energy Transition. Fast Track for RACE for 2030.
- Muhammad, S., Hoffmann, C., 2024. From investment to impact: the role of green finance and technological innovation on German energy transition. *Renew. Energy* 237, 121665. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0960148124017336>
- Müller, M., Sacco, D., 2024 Jul. 3. Energy Transition: Investment Perspective. CIO Special, Deutsche Bank
- Naturstrom AG, 2024. Historie. [Online]. Available: <https://www.naturstrom.de/ueber-uns/naturstrom-ag/historie> (Accessed: 10-Dec-2024).
- New Power, 2019 Jul. Interview: ANESCO - Will a future supply offer be energy in a box? New Power (125). Accessed from <https://www.newpower.info/wp-content/uploads/2019/07/New-Power-July-2019-business-models.pdf>
- Next Kraftwerke, 2023. Redispatch 2.0 – ein Drama in drei Akten. [Online]. Available: <https://www.next-kraftwerke.de/energie-blog/redispatch-2-0-uebergangsloesung> (Accessed on 13 May 2025).
- Nicolli, F., Vona, F., 2019. Energy market liberalization and renewable energy policies in OECD countries. *Energy Policy* 128, 853–867. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0301421519300187>
- Pant, P., Hamacher, T., Perić, V.S., 2024. Ancillary services in Germany: present, future and the role of battery storage systems. *Author's Preprints*.
- Pant, P., Terzija, V., Hamacher, T., Perić, V.S., 2025. Grid agnostic droop control strategy for damping Restoration and optimal reactive power-sharing. *IEEE Open J. Power Electron.* 6, 613–629.
- Pedro, A., Krutnik, M., Yadack, V.M., Pereira, L., Morais, H., 2023. Opportunities and challenges for small-scale flexibility in European electricity markets. *Utilities Policy* 80, 101477.
- Pepermans, G., 2019. European energy market liberalization: experiences and challenges. *Int. J. Econ. Policy Stud.* 13 (1), 3–26.
- PitchBook Data, Inc, 2023 Jun. Q1 2023 clean energy report. [Online]. Available: <https://pitchbook.com/> (Accessed: 10-Dec-2024).
- Plank, J., Dobliger, C., 2018. The firm-level innovation impact of public R&D funding: evidence from the German renewable energy sector. *Energy Policy* 113, 430–438. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0301421517307747>
- Poudineh, R., Apostolopoulou, D., 2024. Powering the future: the energy storage in Tomorrow's electricity markets. In: *Oxford Energy Forum: A Quarterly Journal for Debating Energy Issues and Policies*, no. 140. Oxford: The Oxford Institute for Energy Studies.
- Prashantgrid. n.d. Energy Ventures Dataset. https://github.com/Prashantgrid/Energy_Ventures_dataset
- PV Tech, 2021. BayWa r.E. profits from renewables megatrend to post record financial results. [Online]. Available: <https://www.pv-tech.org/baywa-r-e-profits-from-renewables-megatrend-to-post-record-financial-results>
- Pérez-Arriaga, I.J., Bharatkumar, A., 2016. A framework for redesigning distribution network use of system charges under high penetration of distributed energy resources: new principles for new problems. *Econ. Energy Environ. Policy* 5 (1).

- Qvantum, 2023 Jun. Ambitious heat pump company opens German subsidiary, led by industry veteran, Press Release, [Online]. Available: <https://www.qvantum.com/ambitious-heat-pump-company-opens-#german-subsidiary-led-by-industry-veteran/>
- Rack, Y., 2020 Jul. RWE closes asset swap with E.ON in final step of German power sector revamp. [Online]. Available: <https://www.spglobal.com> (Accessed: 2-Dec-2024).
- Radke, B., 2024. Enpal doubles revenue in financial Year 2023. <https://corporate.enpal.com/pressemitteilungen/enpal-doubles-revenue-in-financial-year-2023>
- Rechsteiner, R., 2021. German energy transition (energiewende) and what politicians can learn for environmental and climate policy. *Clean Technol. Environ. Policy* 23, 305–342.
- Regelleistung.net, 2024 Feb. Präqualifizierte Leistung in Deutschland. Regelleistung.net, Tech. Rep. [Online]. Available: <https://www.regelleistung.net/xspproxy/api/staticfiles/regelleistung/startseite/pq-leistung-in%20deutschland.pdf> (Accessed: 10-Dec-2024)
- Robotron Datenbank-Software GmbH, 2023. CLS management – smart grid control in low-voltage grids (accessed: 2025-May-14. [Online]. Available: <https://www.robotron.de/en/solutions/energy/cls-management>
- Roka, A., Schneider, A., Warack, M., 2013. Polarstern energy – sustainable change led by innovative entrepreneurs. Oikos International, St. Gallen, Switzerland, Case Study 13-ESG-002. Available through the oikos Case Collection. Year of publication is approximate based on typical case development cycles and metadata.
- Rotaru, D.V., 2013. [Online]. Available: A glance at the European energy market liberalization. CES Working Papers 5 (1), 100–110. <https://ideas.repec.org/a/jes/wpaper/y2013v5i1p100-110.html>
- Ruester, S., Schwenen, S., Batlle, C., Pérez-Arriaga, I.J., 2014. From distribution networks to smart distribution systems: rethinking the regulation of European electricity dsos. *Utilities Policy* 31, 229–237.
- Rövekamp, P., Schöpf, M., Wagon, F., Weibelzahl, M., Fridgen, G., 2021. Renewable electricity business models in a post feed-in tariff era. *Energy* 216, 119228. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0360544220323355>
- Samseth, E., Stockhausen, F., Veillard, X., Weiss, A., 2021 Oct. Five trends reshaping European power markets. McKinsey & Company, Tech. Rep. (accessed: 10-Dec-2024. [Online]. Available: <https://www.mckinsey.com>
- Sareen, S., Haarstad, H., 2022. Distributed energy and the challenge of regulation: the German experience with prosumer aggregation. *Sustainability* 14 (13), 7545.
- Schmid-Petri, H., Elschner, S.G., 2024. [Online]. Available: Transitionalists, traditionalists or pioneers? How German municipal energy companies are responding to the national energy transition. *Energy Res. Soc. Sci.* 109, 103431. <https://www.sciencedirect.com/science/article/pii/S2214629624000227>
- Schwarz, H., Cai, X., 2017. Integration of renewable energies, flexible loads and storages into the German power grid: actual situation in German change of power system. *Front. Energy* 11, 107–118.
- SFV, 2012. EEG 2009 inkl. Änderungen EEG 2012 bis 31.3.2012 Einspeisevergütung. https://www.sfv.de/artikel/eeg_2009_inkl_aenderungen_eeg_2012_bis_3132012_einspeiseverguetung. (Accessed: Dec. 22, 2024).
- Sievers, L., Breitschopf, B., Pfaff, M., Schaffer, A., 2019. Macroeconomic impact of the German energy transition and its distribution by sectors and regions. *Ecol. Econ.* 160, 191–204. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0921800918310073>
- Singh, M., Jiao, J., Klobasa, M., Frietsch, R., 2021. Making energy-transition headway: a data driven assessment of German energy startups. *Sustain. Energy Technol. Assess.* 47, 101322. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2213138821003325>
- Sousa, T., Soares, T., Pinson, P., Moret, F., Baroche, T., Sorin, E., 2019. Peer-to-peer and community-based markets: a comprehensive review. *Renew. Sustain. Energy Rev.* 104, 367–378.
- Start Up Energy Transition Initiative, 2024. SET100 list 2024. German Energy Agency (dena) and World Energy Council, <https://www.startup-energy-transition.com/app/uploads/2024/03/SET100-List-2024-2.pdf>
- Stede, J., Arnold, K., Dufter, C., Holtz, G., von Roon, S., Richstein, J.C., 2020. The role of aggregators in facilitating industrial demand response: evidence from Germany. *Energy Policy* 147, 111893. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S030142152030608X>
- Sungrow Power Supply Co., Ltd, 2025 Apr. Stem-cell grid-forming Tech 2.0 white paper. Sungrow. White Paper. [Online]. Available: <https://en.sungrowpower.com/upload/file/20250407/Stem-cell%20Grid-forming%20Tech%202.0%20White%20Paper.pdf>. (Accessed: 2025-May-12).
- Sympower, 2022. Sympower raises €21 million to support European energy transition. [Online]. Available: <https://www.sympower.net/news/sympower-raises-21-million> (accessed: 2025-May-14).
- Tesvolt AG, 2025 5 February. New TESVOLT subsidiary paves the way for profitable energy trading with commercial storage systems. Press release, Feb. 2025, press release available at <https://www.tesvolt.com/en/press/new-tesvolt-subsidiary-paves-the-way-for-profitable-energy-trading-with-commercial-storage-systems.html>. (Accessed on 2025-05-13). (Content indicates: “Start-up TESVOLT Energy debuts at E-world”. Date referenced in text: Berlin, Wittenberg)
- TESVOLT AG, 2025. TESVOLT Projects – Full-Service für kommunale Batteriepark. [Online]. Available: <https://www.tesvolt-projects.com/de/> <https://www.tesvolt-projects.com/de/> (Accessed: 2025-May-16).
- Thermondo GmbH, 2025. Thermondo Fördermittelservice: Bis 70 für Wärmepumpe. Copyright 2013-2025. [Online]. Available: <https://www.thermondo.de/leistungen/komplettpaket-waermepumpe/foerdermittelservice/>. (Accessed on 13 May 2025).
- TransnetBW GmbH, TransnetBW GmbH, 2024 Dec. TransnetBW nimmt virtuellen Kraftwerksblock von sonnen in Betrieb. TransnetBW GmbH, Press Release, [Online]. Available: https://www.transnetbw.de/_Resources/Persistent/f/2/3/f/f23ffa874fbac54db9ce70156255be627761ff99/2024-12-12_TransnetBW_sonnen_Pressemitteilung.pdf
- TWAICE, 2025. TWAICE rolls out 15+ product enhancements to help BESS operators turn complexity into revenue: new capabilities empower operators to navigate complexity, set internal benchmarks, and drive stronger financial outcomes. *GlobeNewswire*. [Online]. Available: <https://www.globenewswire.com/news-release/2025/05/06/3075135/0/en/TWAICE-Rolls-Out-15-Product-Enhancements-to-Help-BESS-Operators-Turn-Complexity-into-Revenue.html>. (Accessed: 2025-May-22).
- Umweltbundesamt, 2024. Zeitreihen zur Entwicklung der erneuerbaren Energien. [Online]. Available: <https://www.umweltbundesamt.de/dokument/zeitreihen-zur-entwicklung-der-erneuerbaren> (Accessed: 3-Dec-2024).
- VDE Association for Electrical, Electronic and Information Technologies, 2018. VDE-AR-N 4105: 2018-11 - technical requirements for the connection and operation of Generating Plants in the low-voltage Network. Available at: <https://www.vde.com>
- VDE Association for Electrical, Electronic and Information Technologies, 2019. VDE-AR-N 4100: 2019-04 - technical requirements for the connection and operation of customer installations to the Low-voltage network (TAR). Available at: <https://www.vde.com>
- Verheyen, S., Graf, A., 2025. Boosting the clean heat market. A policy for guiding the transition of the EU heating industry.
- Vienna University of Technology, 2004 Sep. Deriving optimal promotion strategies for increasing the share of RES-E in a dynamic European electricity market. Green-X Project, DG Research, Fifth RTD Framework Programme. Tech. Rep. Duration: Oct. 2002 – Sep. 2004, Available online: <http://www.green-x.at>
- Wigand, F., Förster, S., Amazo, A., Tiedemann, S., 2016 Jun. Auctions for renewable energy support: lessons learnt from international experiences. AURES Project. Deliverable D4.2 [Online]. Available: https://aures2project.eu/wp-content/uploads/2020/03/AURES_WP4_Synthesis_Report.pdf (Accessed: 2025-May-14).
- World Economic Forum, 2025 Jan. Accelerating the energy transition: unpacking the business and economic cases. [Online]. Available: https://reports.weforum.org/docs/WEF_Accelerating_the_Energy_Transition_2025.pdf
- Xia-Bauer, C., Vondung, F., Thomas, S., Moser, R., 2022. Business model innovations for renewable energy prosumer development in Germany. *Sustainability* 14 (13), 7545. [Online]. Available: <https://www.mdpi.com/2071-1050/14/13/7545>
- Zou, Y., Xu, Y., 2025. DER-Inverter based reactive power ancillary service for supporting peer-to-peer transactive energy trading in distribution networks. *IEEE Trans. Power Syst.* 40 (1), 753–764.
- Zentrum für Sonnenenergie- und Wasserstoff-Forschung Baden-Württemberg (ZSW), 2024. Economic impetus from renewable energy: figures and data on the expansion of renewable energy as an economic factor. <https://www.bmwk.de/Redaktion/EN/Downloads/Energie/kurzdokumentation-wirtschaftl-impulse-ee-2024-eng.pdf>